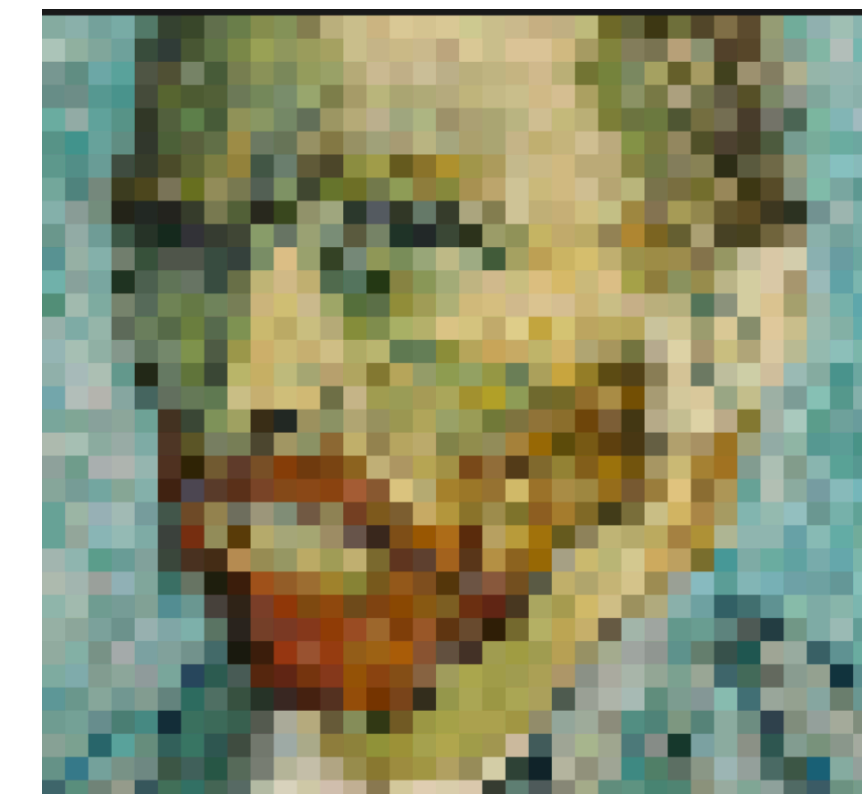
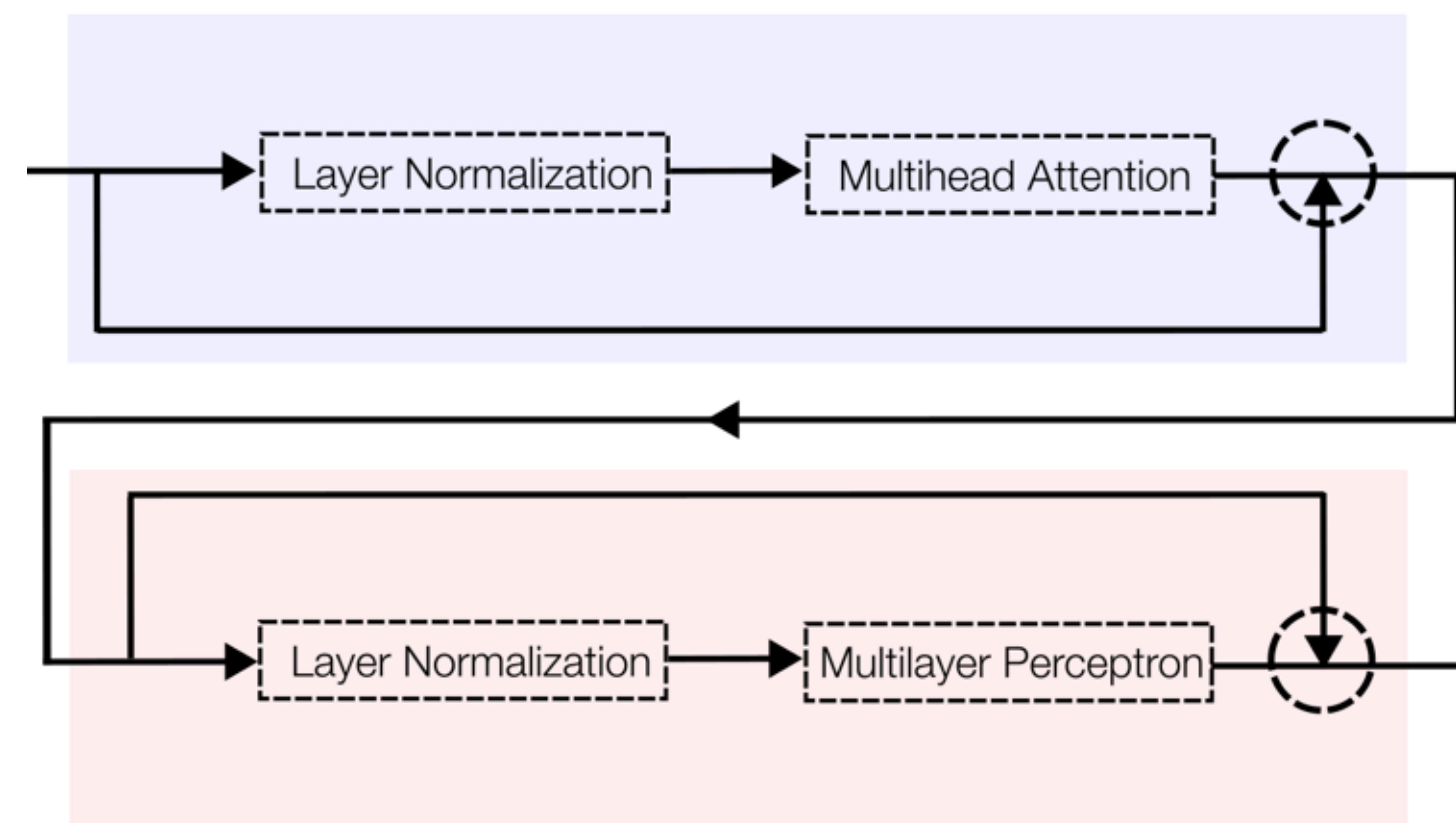
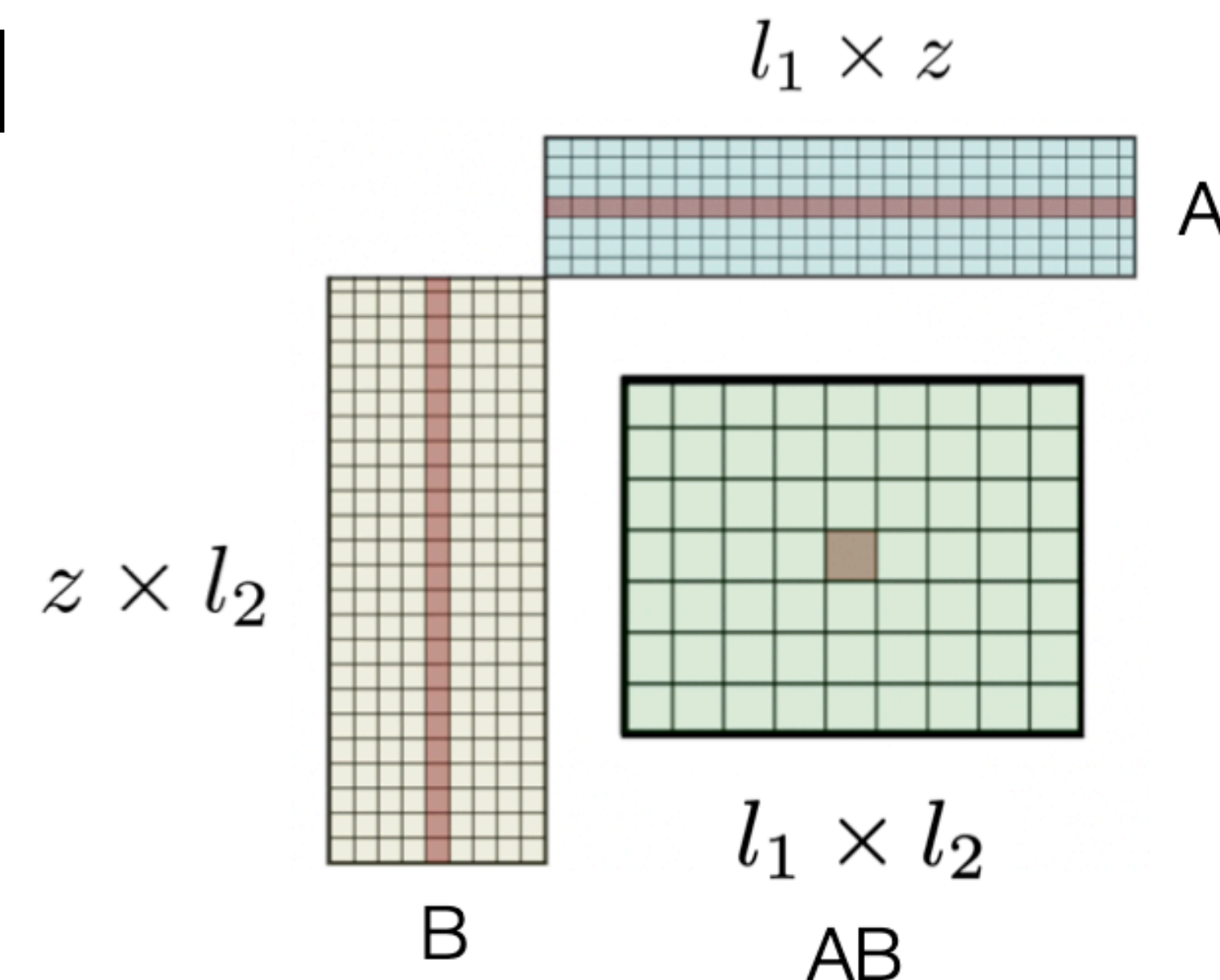
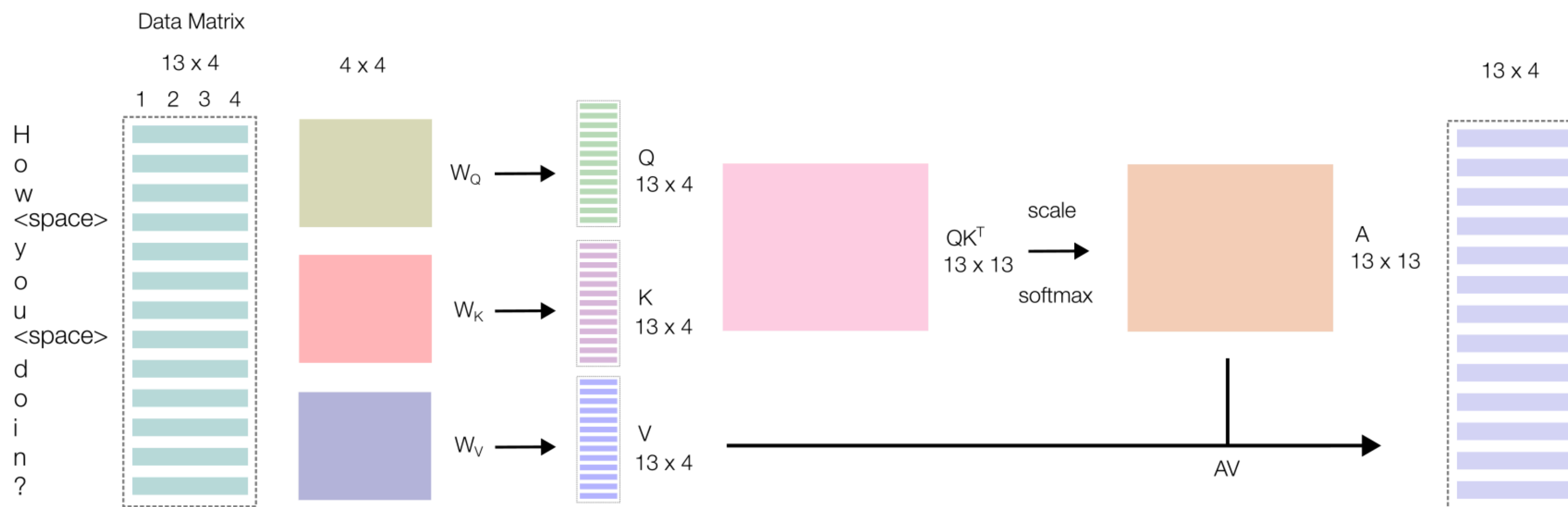


What authority surfeits on would relieve us
 What authority surfeits on would relieve us
 What authority surfeits on would relieve us
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 What authority surfeits on would relieve us
 What authority surfeits on would relieve us
 What authority surfeits on would relieve us
 What authority surfeits on would relieve us



Build a language model from scratch!

<https://tinyurl.com/257avm9d>



Progress Bar

- Very short intro to machine learning
- Detailed overview of transformers + code

Progress Bar

- Very short intro to machine learning

- Data

- Model

- Training Objective

- Learning Algorithm

Data

Modalities

Text



"Et tu Brute??!!?" *ATGACTA*

Images



Sound



Data table



[neck_length, body_mass, wing_span, ...]
[0.67, 0.083, 1.63, ...]
[1.28, 7.23, 0, ...]

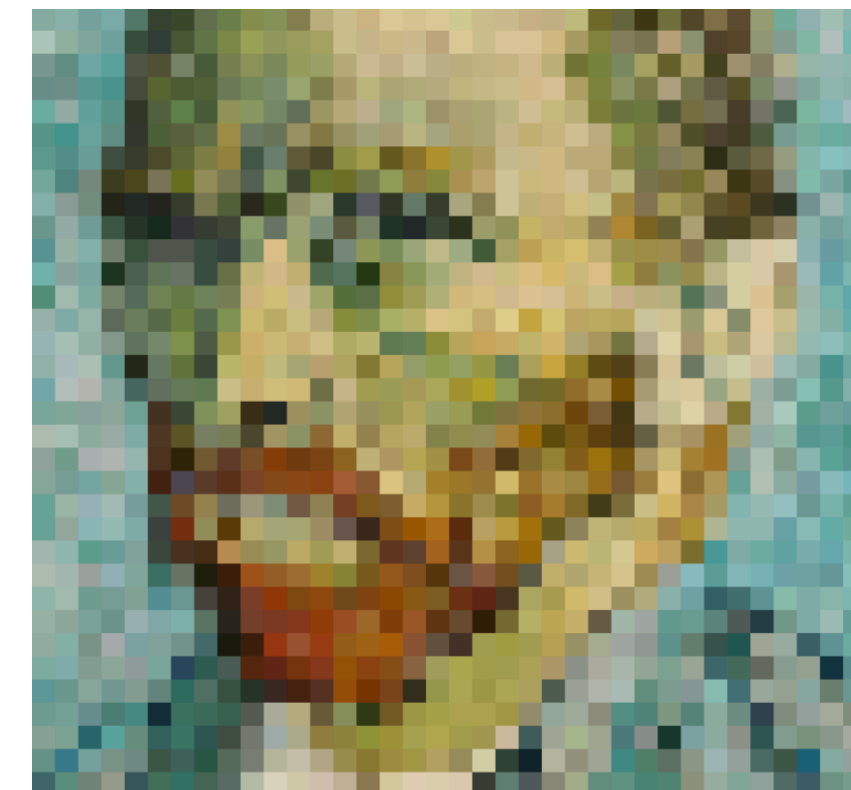
All modalities can be represented
as sequences of numbers...

Vectors

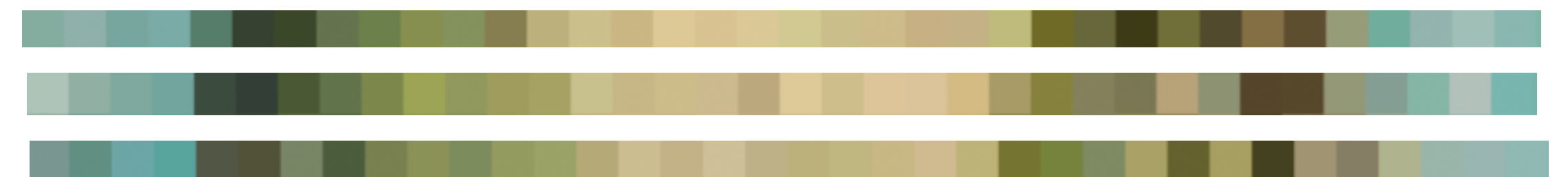
Vectors serve as a representation of the data

		1	2	3	4
1	H				
2	o				
3	w				
4	<space>				
5	y				
6	o				
7	u				
8	<space>				
9	d				
10	o				
11	i				
12	n				
13	?				

Text



Slice the image to get ...



... a sequence of pixels

Progress Bar

- Very short intro to machine learning

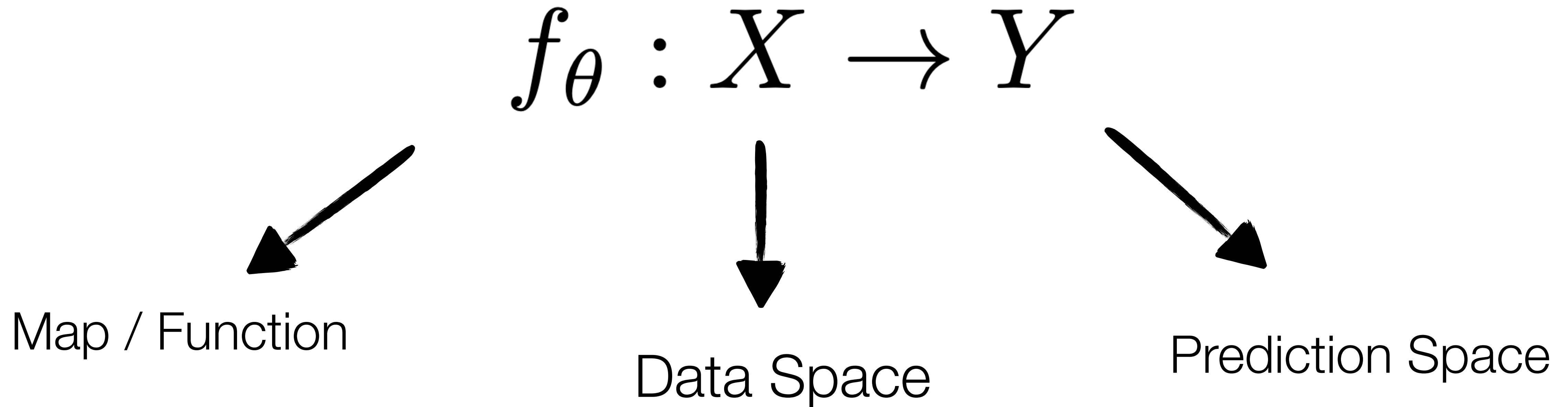
- ~~Data~~ *Practically anything can be turned into a vector*
- Model
- Training Objective
- Learning Algorithm

Model

A useful model takes a datapoint as input and outputs a sensible prediction.

This mapping is learned over the course of training.

$$f_{\theta} : X \rightarrow Y$$



Map / Function

Data Space

Prediction Space

Neural networks can be expressed as a composition of simple primitives

$$f_{\theta} : X \rightarrow Y$$

The specific arrangement of the primitives is what defines the architecture.

Linear Operation or Matrix multiplication	}	Multilayer Perceptron (MLP)
Non-linear transform		

Linearity

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$$

$$A(\alpha\mathbf{x}) = \alpha A\mathbf{x}$$

Matrix

$$\begin{array}{c}
 1 \\
 2 \\
 3 \\
 \vdots \\
 m
 \end{array}
 \begin{array}{c}
 1 \quad 2 \quad \cdots \quad n
 \end{array}
 \begin{bmatrix}
 a_{11} & a_{12} & \cdots & a_{1n} \\
 a_{21} & a_{22} & \cdots & a_{2n} \\
 a_{31} & a_{32} & \cdots & a_{3n} \\
 \vdots & \vdots & & \vdots \\
 a_{m1} & a_{m2} & \cdots & a_{mn}
 \end{bmatrix}$$

Each entry is a tunable parameter

Linear Operations

Matrices: A convenient way to manipulate vectors

$$\underbrace{\begin{bmatrix} \\ \\ \end{bmatrix}}_{m \times n \text{ matrix}} \times \underbrace{\begin{bmatrix} \\ \end{bmatrix}}_{n\text{-dimensional vector}} = \underbrace{\begin{bmatrix} \\ \end{bmatrix}}_{m\text{-dimensional vector}}$$

Matrices can also act on other matrices

$$\underbrace{\begin{bmatrix} \\ \\ \end{bmatrix}}_{m \times n} \times \underbrace{\begin{bmatrix} \\ \end{bmatrix}}_{n \times p} = \underbrace{\begin{bmatrix} \\ \end{bmatrix}}_{m \times p}$$

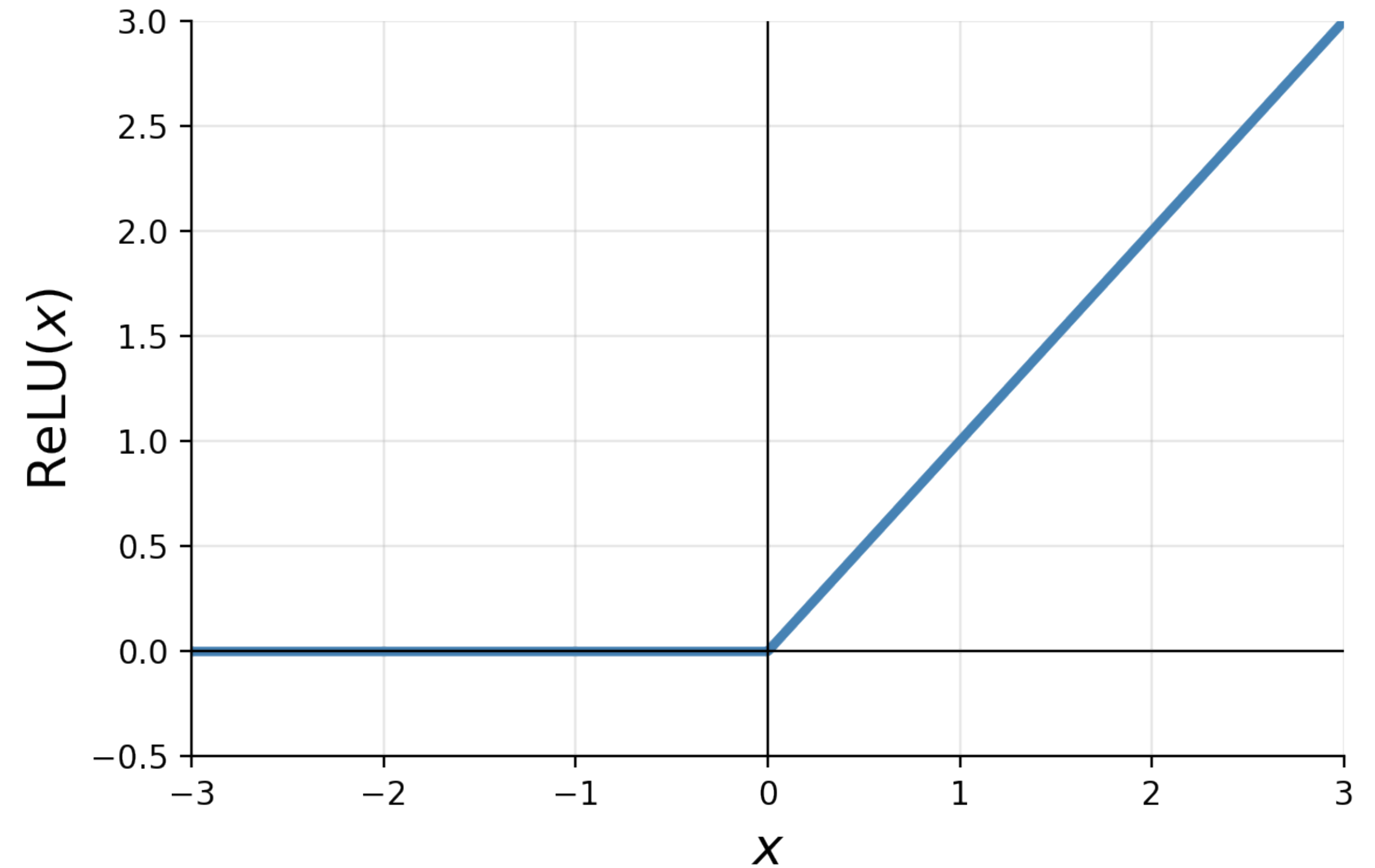
Linear operations on their own can be quite limited...
... since they can collapse to a single operation.

$$ABCD(x) = Z(x)$$

We need something more to add expressivity...

Non-linear Operations

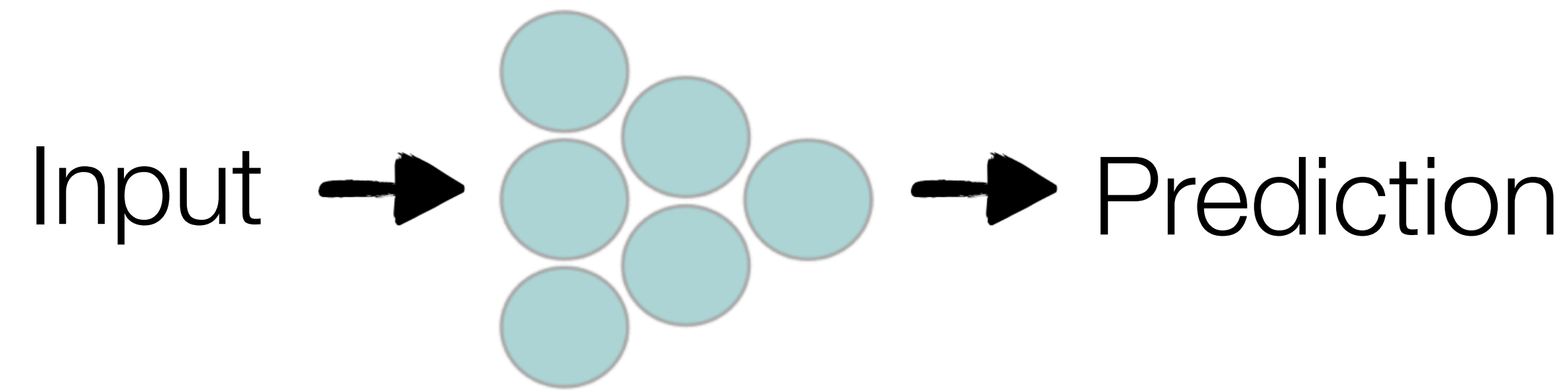
$$\text{ReLU}(x) = \max(0, x)$$



$[0.67, -0.083, 1.63, -2.68, 1.31, \dots]$ $\xrightarrow{\text{ReLU}}$ $[0.67, 0, 1.63, 0, 1.31, \dots]$

The operation is applied point-wise.

Multilayer Perceptron (MLP)



ReLU non-linearity kills negative entries

Data vector

$A - \text{ReLU} - B - \text{ReLU} - C$

Prediction

Linear Operation: Matrix multiplication

Progress Bar

- Very short intro to machine learning

- ~~Data~~ *Practically anything can be turned into a vector*
- ~~Model~~ *Neural networks are composed of simple primitives*
- Training Objective
- Learning Algorithm

Training Objective

Loss functions measure how wrong the model is in its predictions.

$$f_{\theta} : X \rightarrow Y \qquad \mathcal{L}(\theta) = \mathcal{L}(f_{\theta}(x), y)$$

A larger loss implies a larger deviation between the model's behavior and optimal behavior.

Classification

Use model to determine Ostrich / Giraffe from the data vector

[neck_length, body_mass, wing_span, ...]	Species
--	---------

[0.67, 0.83, 0.18, ...]	Ostrich
-------------------------	---------

[1.28, 7.23, 0, ...]	Giraffe
----------------------	---------

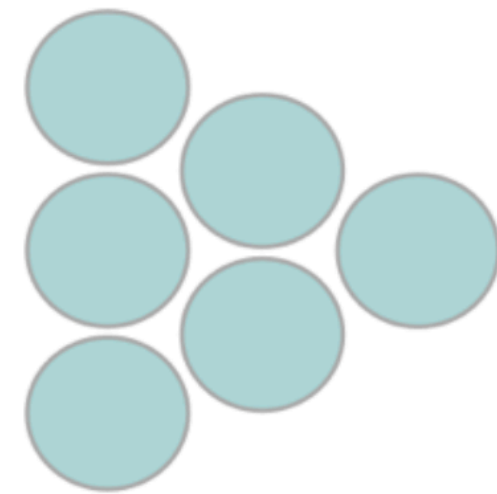
Another primitive

$$\text{softmax}(z_k) = \frac{e^{z_k}}{\sum_j e^{z_j}}$$

MLP + softmax

Input

[0.67, 0.83, 0.18, ...]



Prediction

Ostrich: 0.3 , Giraffe: 0.7

Truth

Ostrich: 1 , Giraffe: 0

$$D_{KL}(y||p) = \sum_k y_k \log \frac{y_k}{p_k} = 1 \cdot \log \frac{1}{0.3} + 0 \cdot \log \frac{0}{0.7} = -\log(0.3) \approx 1.204$$

Progress Bar

- Very short intro to machine learning

- ~~Data~~ *Practically anything can be turned into a vector*

- ~~Model~~ *Neural networks are composed of simple primitives*

- ~~Training Objective~~ *Minimize the loss function*

- Learning Algorithm

Learning Algorithm

$$f_{\theta} : X \rightarrow Y$$

Neural network model f with parameters θ

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (y_i - f_{\theta}(x_i))^2$$

Mean squared loss evaluated over N points

$$\nabla_{\theta} \mathcal{L}(\theta)$$

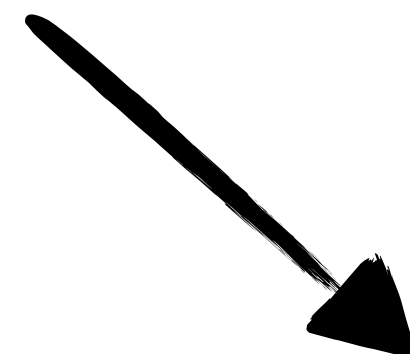
The gradient of the loss points in the direction of the fastest increase in the loss

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}(\theta)$$

Update rule



Updated parameters



Learning rate / Step size

The training and evaluation process...

[...]

[...]

[...]

•
•
•

Training split

Used to learn the mapping between the input and output

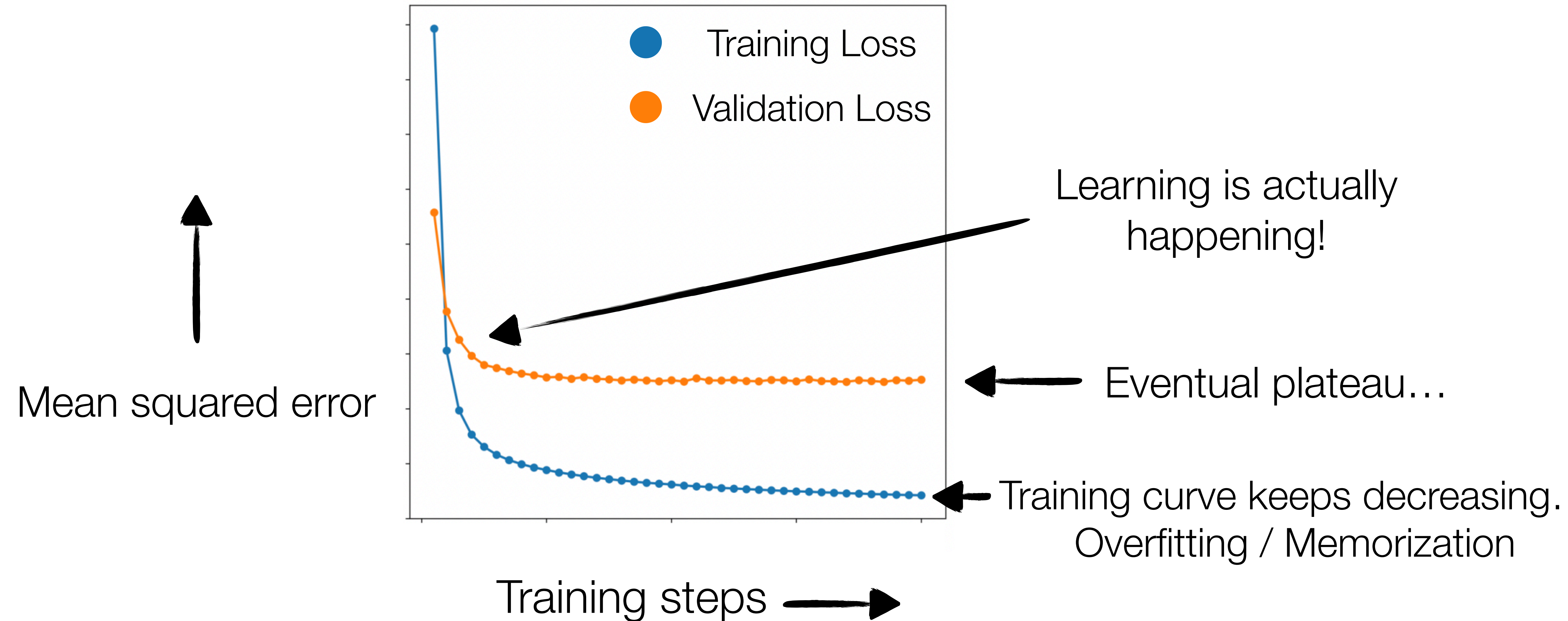
[...]

[...]

Validation split

Used to evaluate if the model generalizes to unseen settings

Loss Curves



Progress Bar

- ~~Very short intro to machine learning~~

- ~~Data~~ *Practically anything can be turned into a vector*

- ~~Model~~ *Neural networks are composed of simple primitives*

- ~~Training Objective~~ *Minimize the loss function*

- ~~Learning Algorithm~~ *Stochastic gradient descent +
bells/whistles*

Progress Bar

- ~~Very short intro to machine learning~~

- Detailed overview of transformers + code

Detailed overview of transformers + code

- Dataset
- Training objective
- Architecture + Training

Dataset

Tiny Shakespeare — Andrej Karpathy

"The Unreasonable Effectiveness of Recurrent Neural Networks"

** Check Colab notebook to explore dataset **

Detailed overview of transformers + code

- Dataset
- Training objective
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Training objective: Next token prediction

Use preceding context to predict the next token...

Preceding context — Training context length —
0 *What authority surfeits on would relieve us*

Training objective: Next token prediction

Use preceding context to predict the next token...

Preceding context — Training context length —

0	<i>What authority surfeits on would relieve us</i>
1	<i>What authority surfeits on would relieve us</i>

Training objective: Next token prediction

Use preceding context to predict the next token...

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2	<i>What authority surfeits on would relieve us</i>
3	<i>What authority surfeits on would relieve us</i>
4	<i>What authority surfeits on would relieve us</i>

Training objective: Next token prediction

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Preceding context ————— Training context length —————

0	<i>What authority surfeits on would relieve us</i>
1	<i>What authority surfeits on would relieve us</i>
2	<i>What authority surfeits on would relieve us</i>
3	<i>What authority surfeits on would relieve us</i>
4	<i>What authority surfeits on would relieve us</i>
5	<i>What_<u>authority</u> surfeits on would relieve us</i>

Training objective: Next token prediction

Use preceding context to predict the next token...

Preceding context ————— Training context length —————

0	<i>What authority surfeits on would relieve us</i>
1	<i>What authority surfeits on would relieve us</i>
2	<i>What authority surfeits on would relieve us</i>
3	<i>What authority surfeits on would relieve us</i>
4	<i>What authority surfeits on would relieve us</i>
5	<i>What_authority surfeits on would relieve us</i>
6	<i>What_authority surfeits on would relieve us</i>

Training objective: Next token prediction

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3	<i>What authority surfeits on would relieve us</i>
4	<i>What authority surfeits on would relieve us</i>
5	<i>What_authority surfeits on would relieve us</i>
6	<i>What_authority surfeits on would relieve us</i>
7	<i>What_authority surfeits on would relieve us</i>

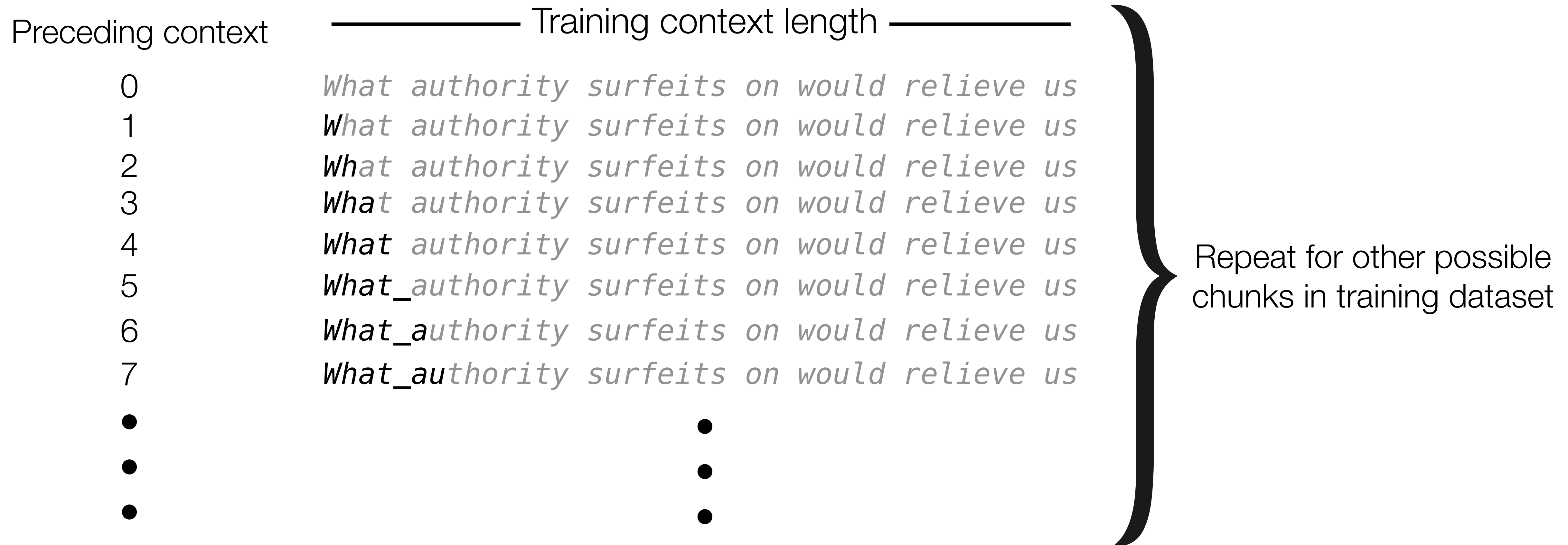
Training objective: Next token prediction

Use preceding context to predict the next token...

Preceding context	Training context length
0	<i>What authority surfeits on would relieve us</i>
1	<i>What authority surfeits on would relieve us</i>
2	<i>What authority surfeits on would relieve us</i>
3	<i>What authority surfeits on would relieve us</i>
4	<i>What authority surfeits on would relieve us</i>
5	<i>What_authority surfeits on would relieve us</i>
6	<i>What_authority surfeits on would relieve us</i>
7	<i>What_authority surfeits on would relieve us</i>
•	•
•	•
•	•

Training objective: Next token prediction

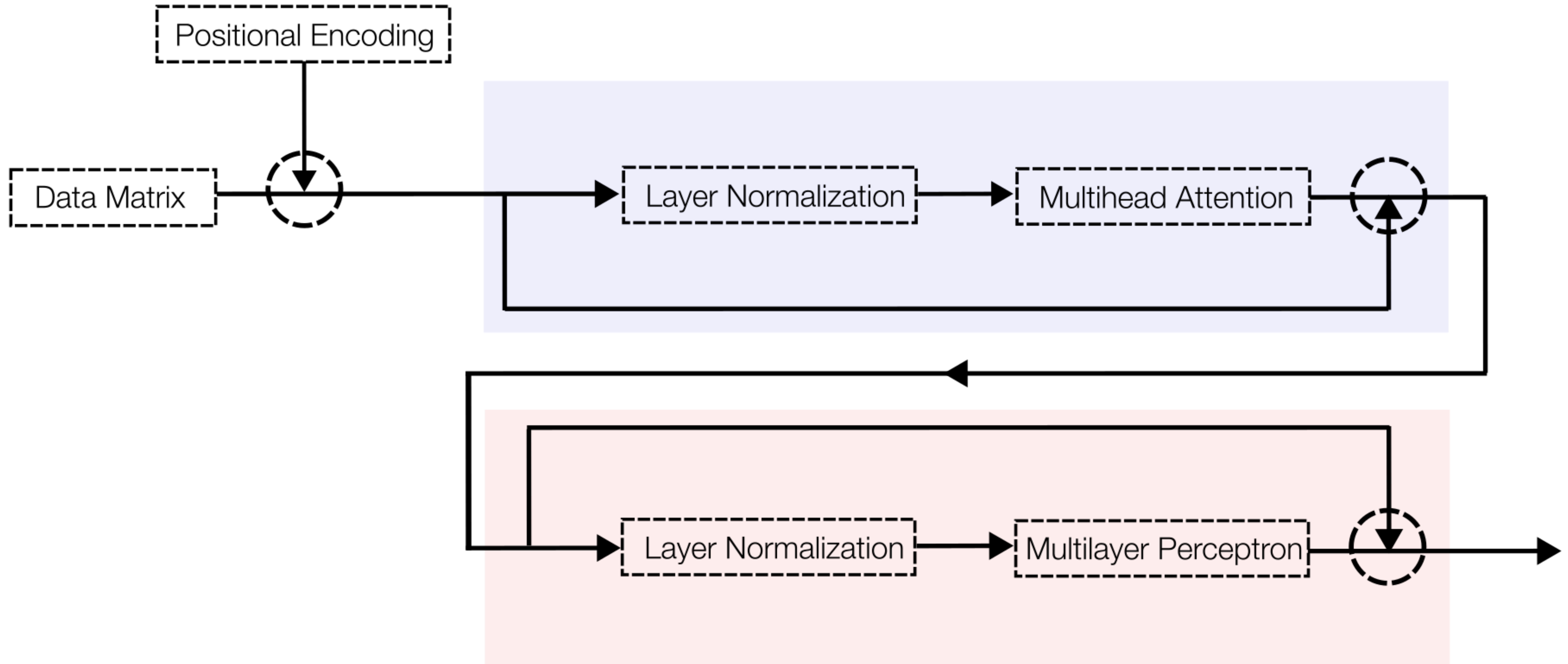
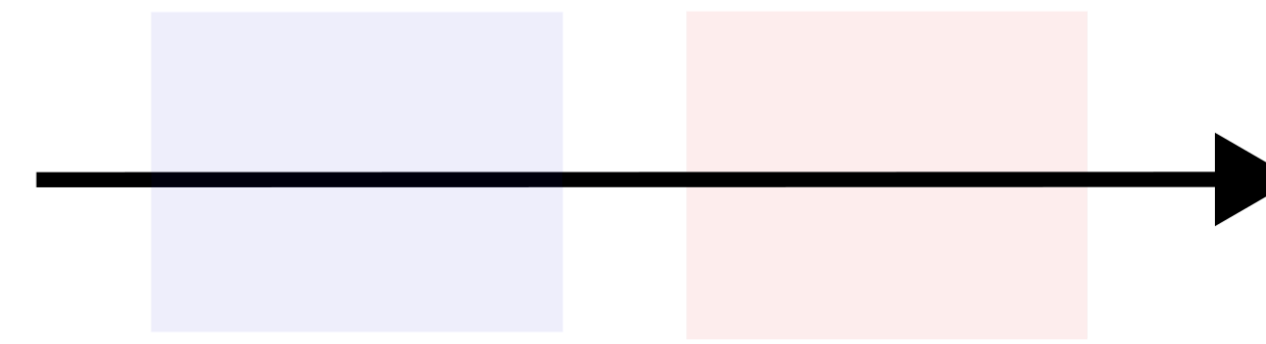
Use preceding context to predict the next token...



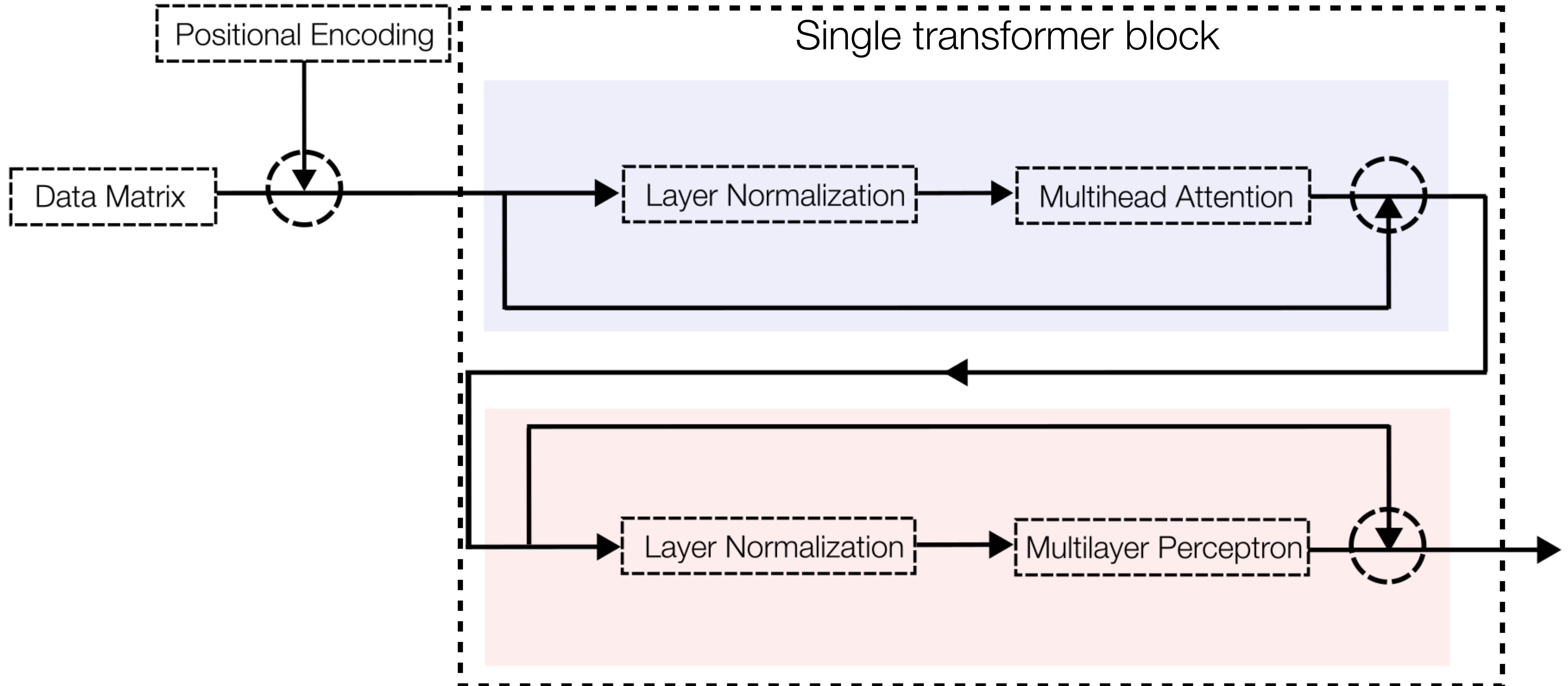
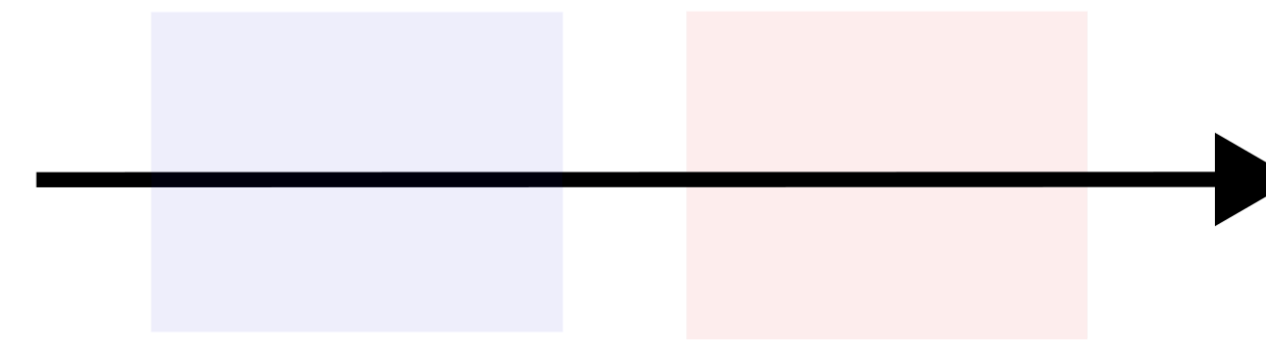
Detailed overview of transformers + code

- Dataset
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Transformer Architecture



Transformer Architecture

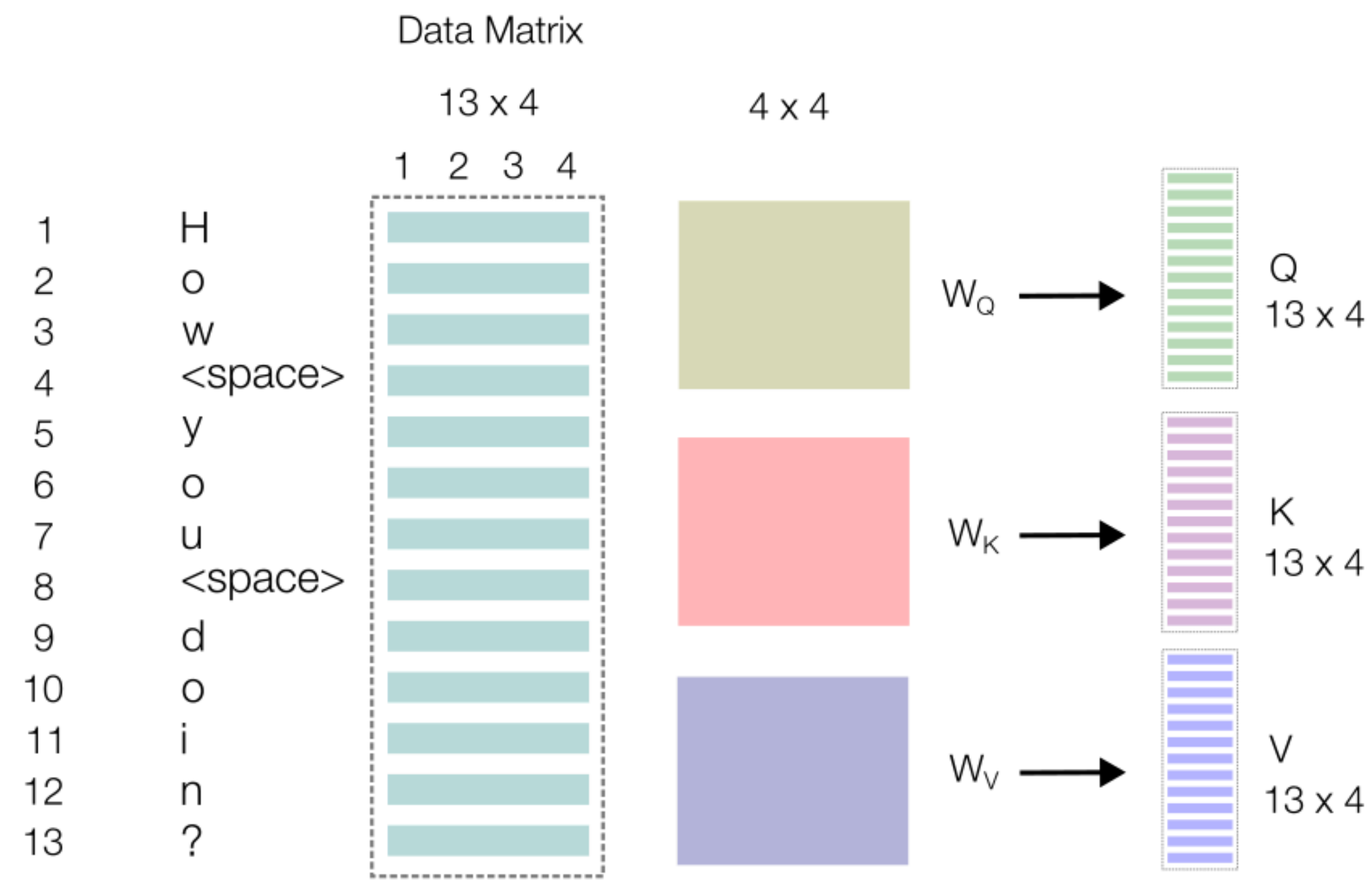


****Single Head Attention****

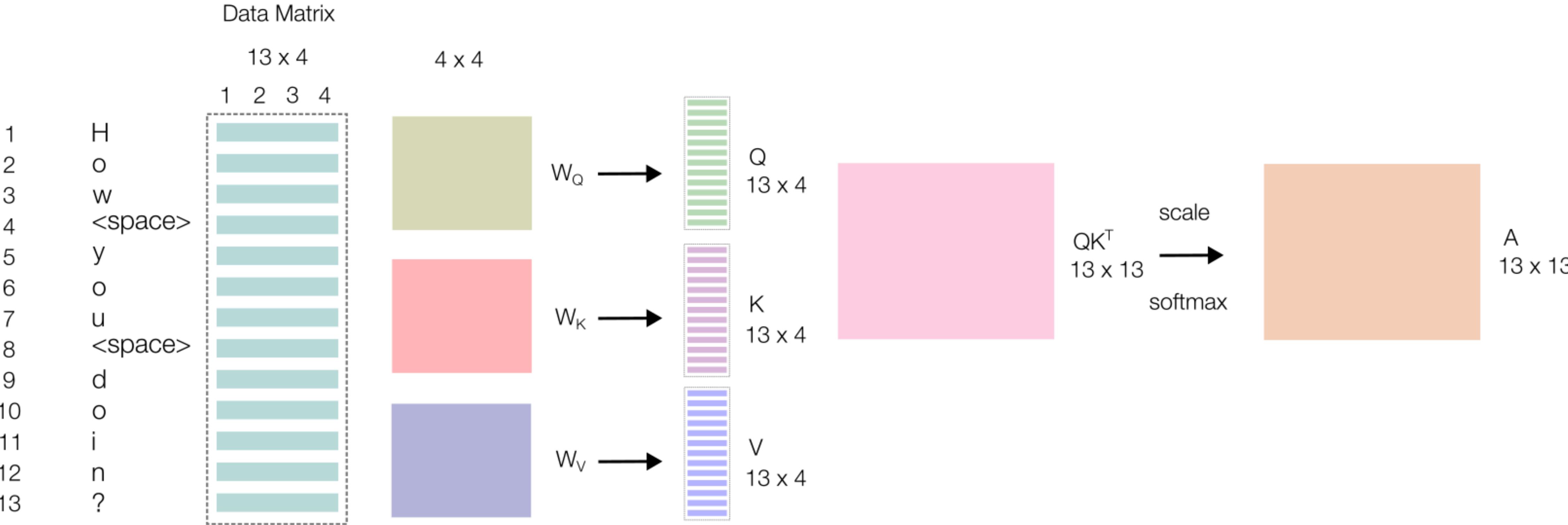
Data Matrix
13 x 4
1 2 3 4

1	H				
2	o				
3	w				
4	<space>				
5	y				
6	o				
7	u				
8	<space>				
9	d				
10	o				
11	i				
12	n				
13	?				

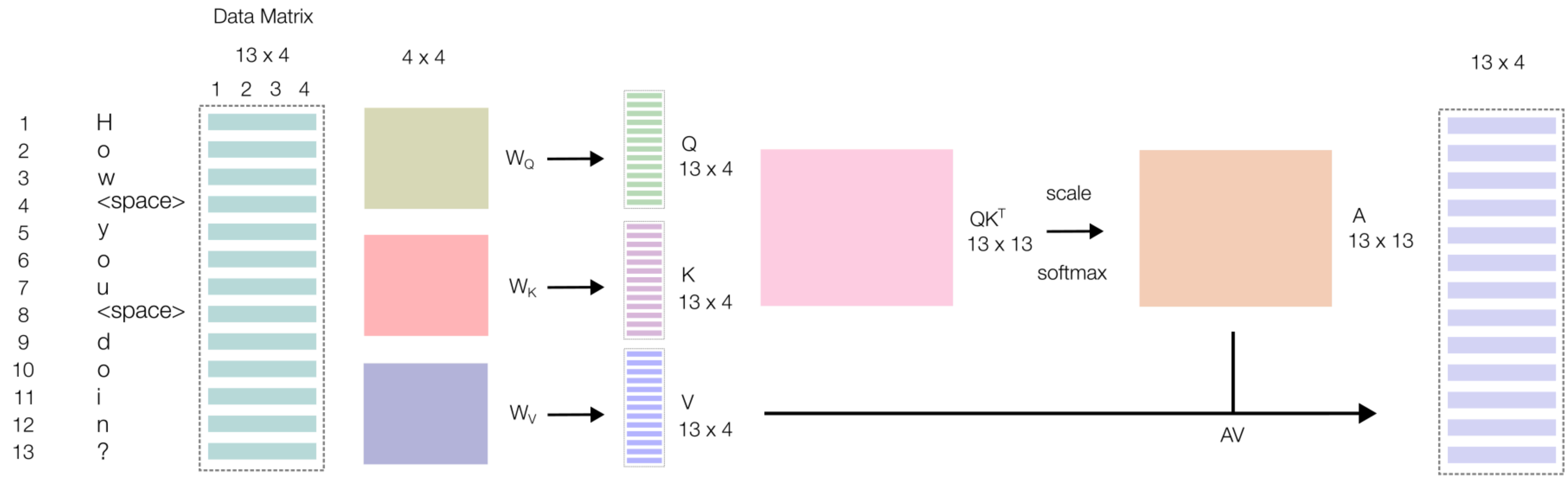
****Single Head Attention****



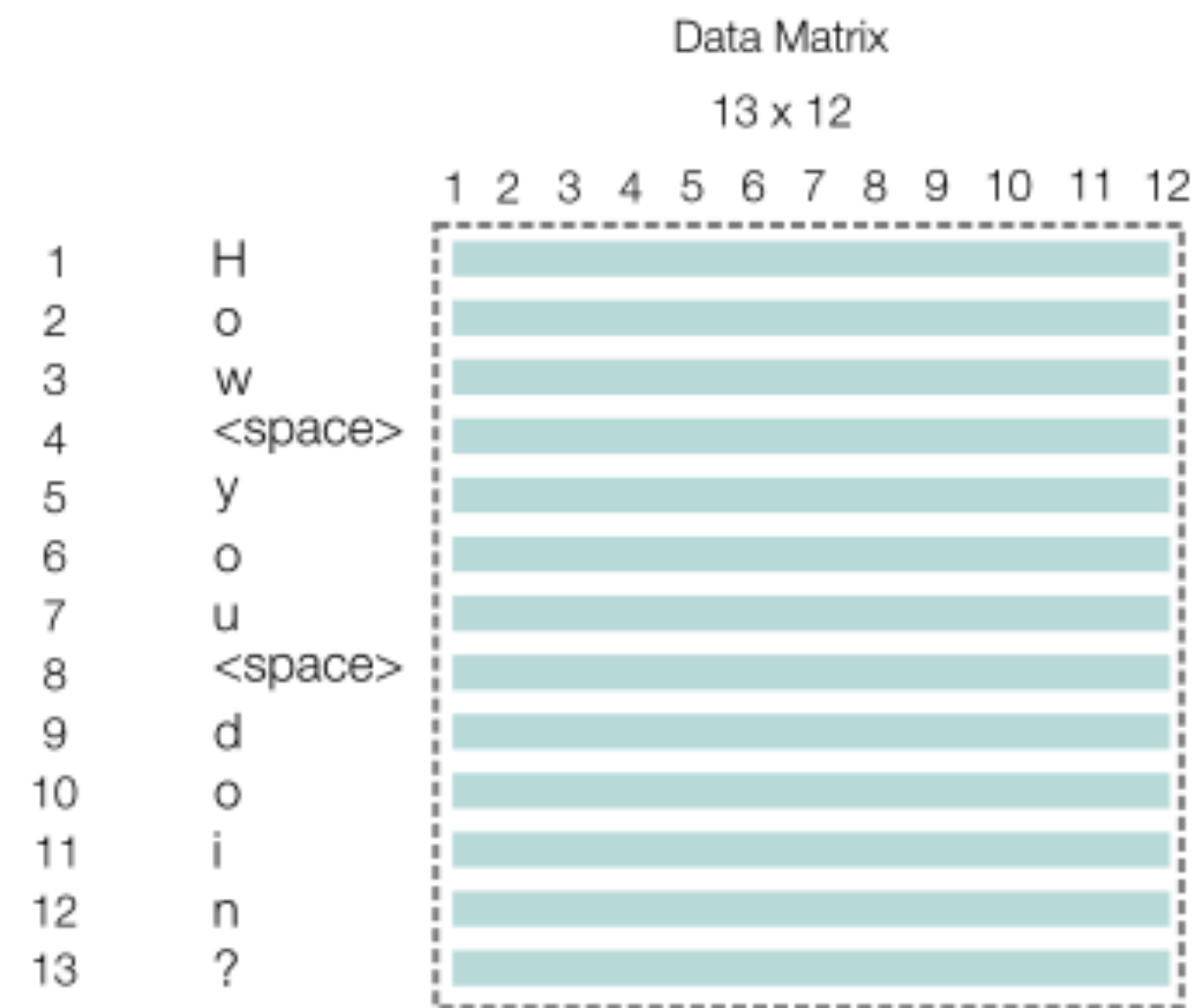
****Single Head Attention****



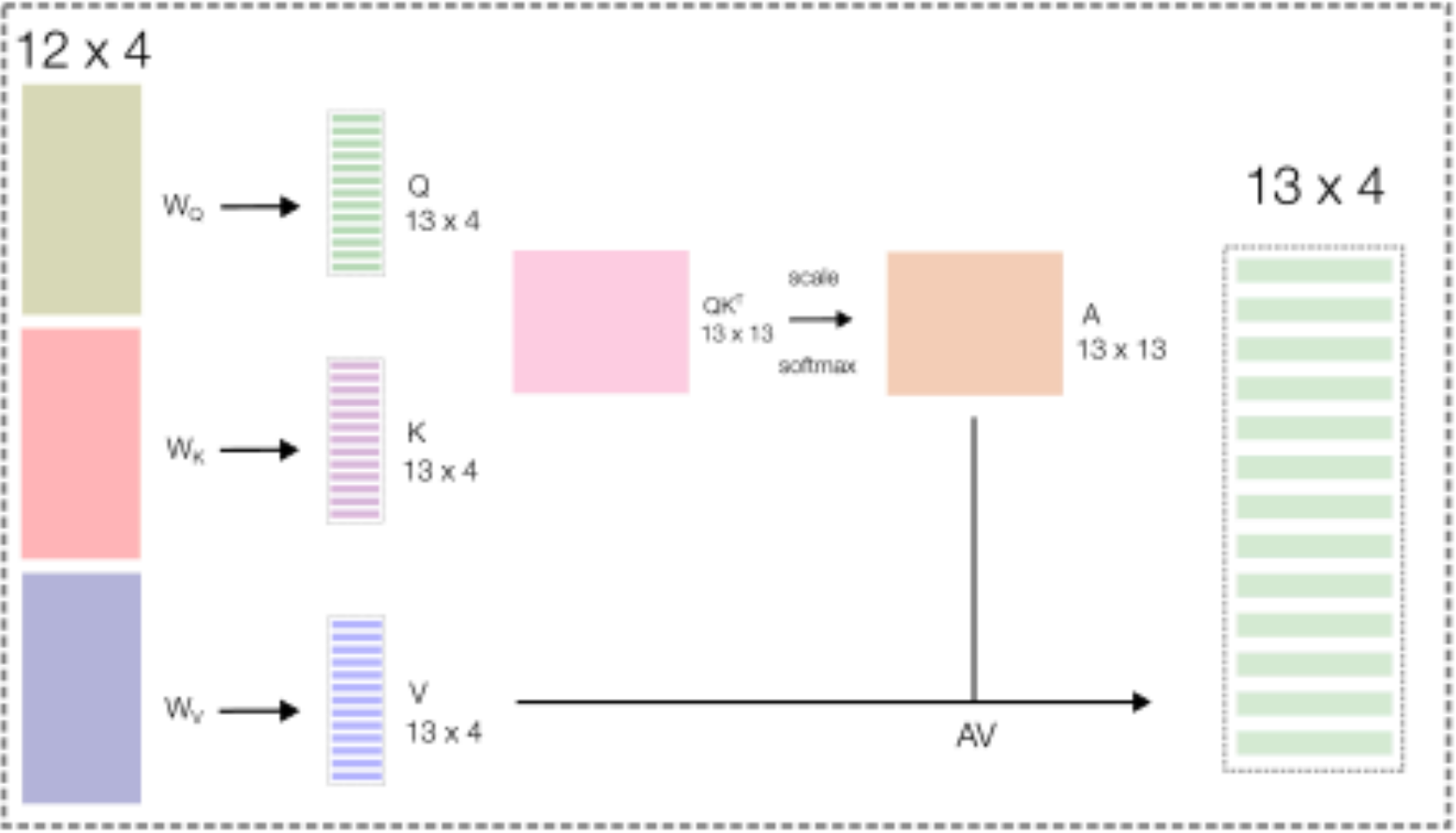
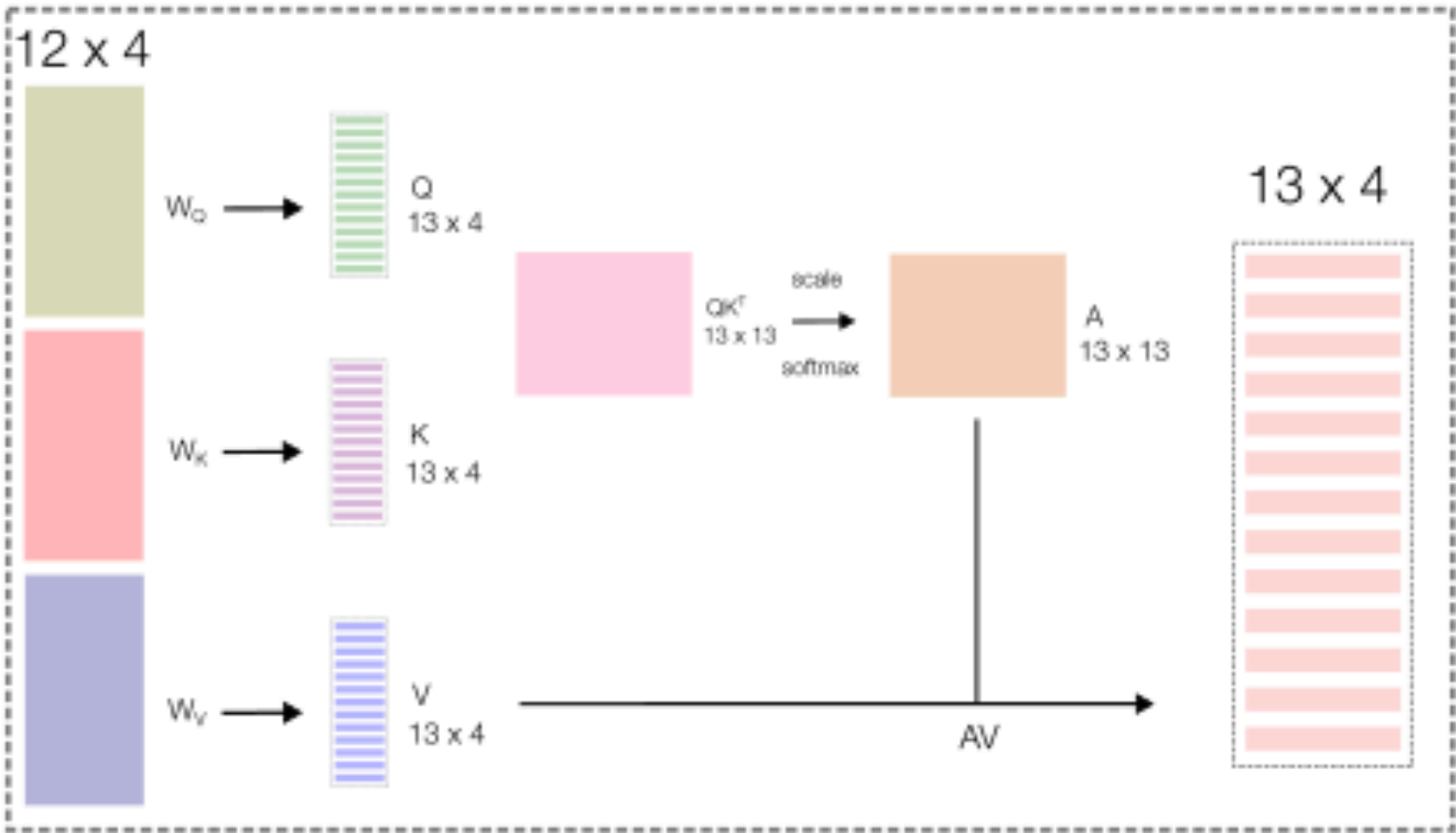
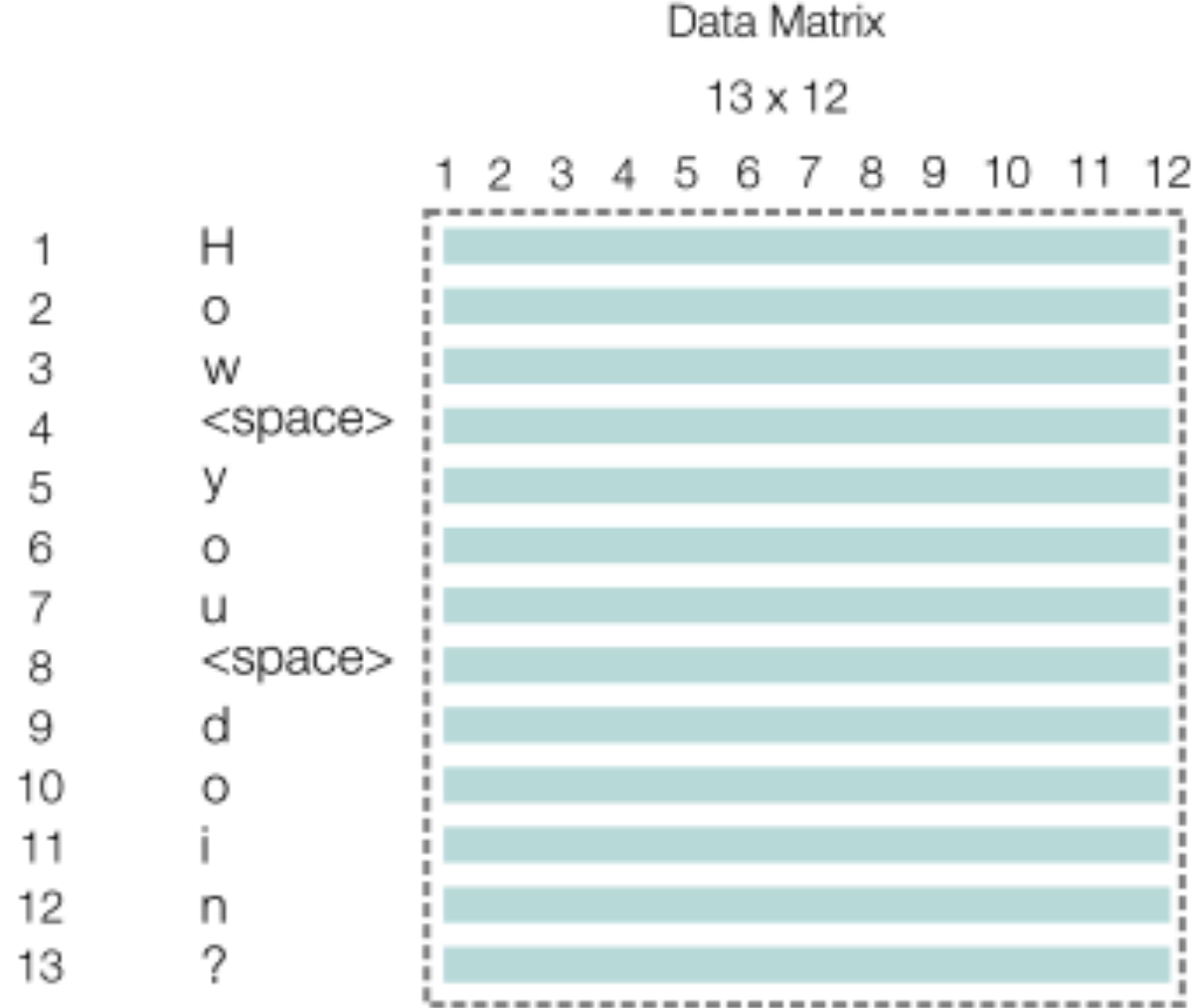
****Single Head Attention****



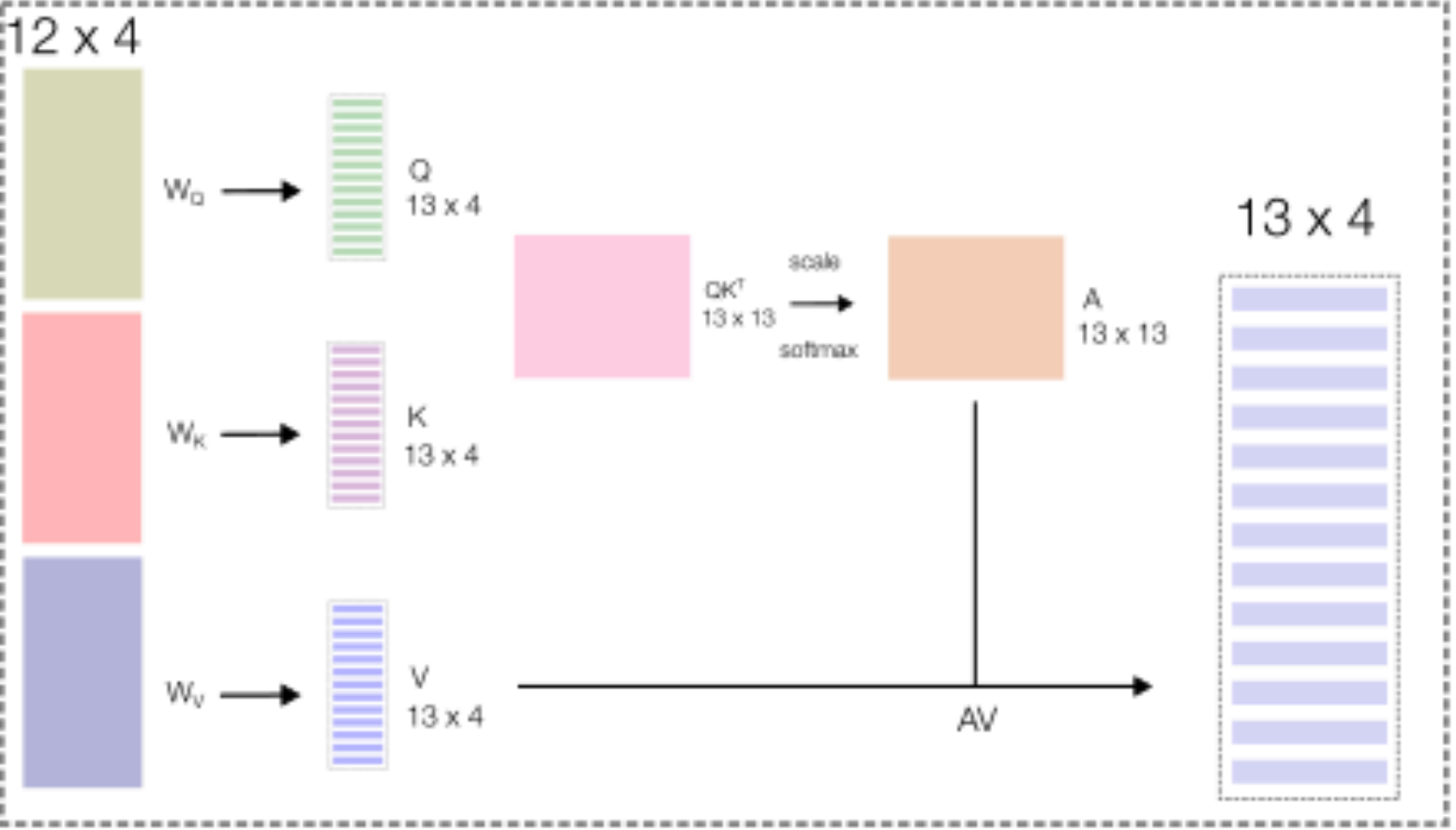
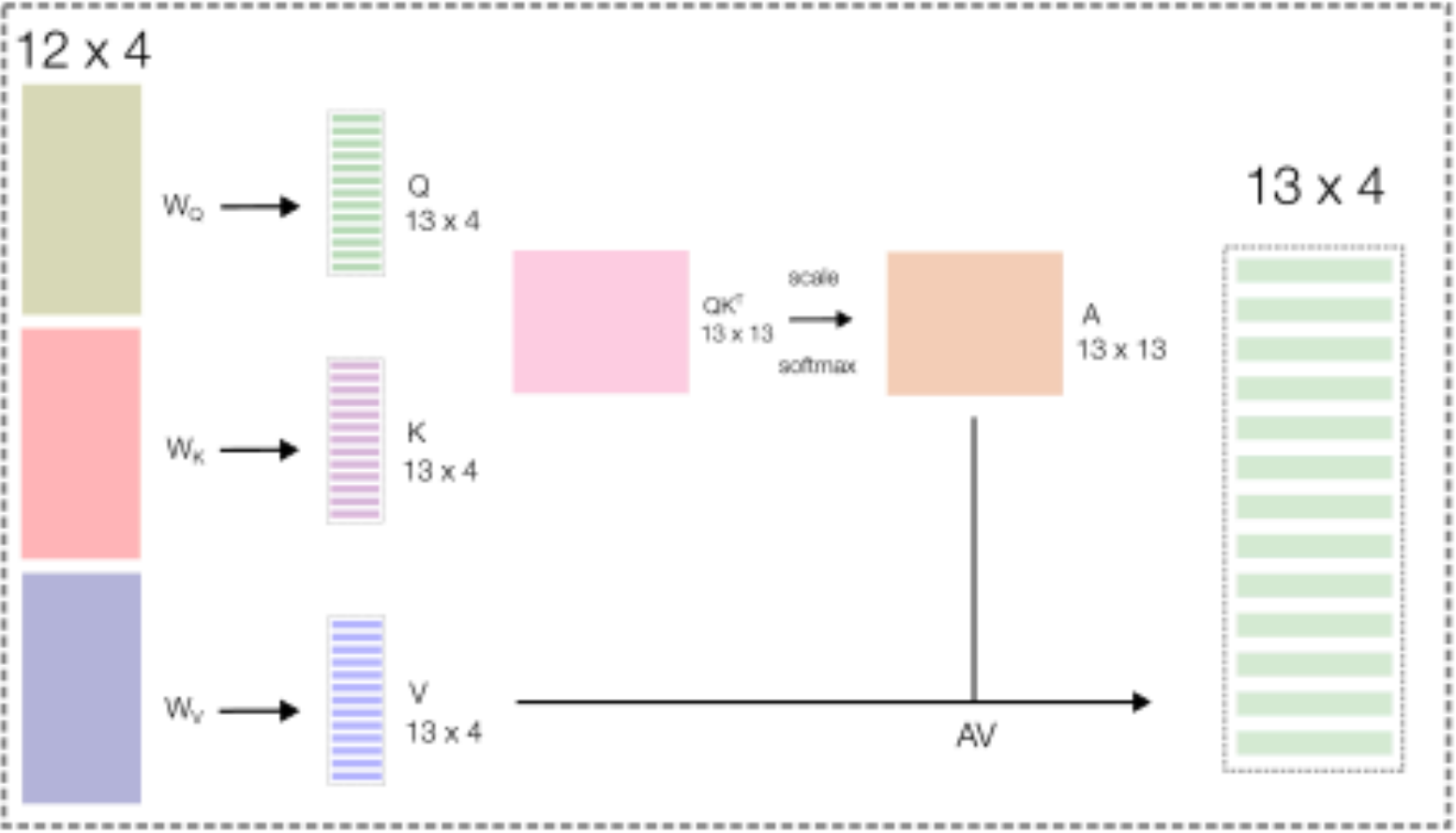
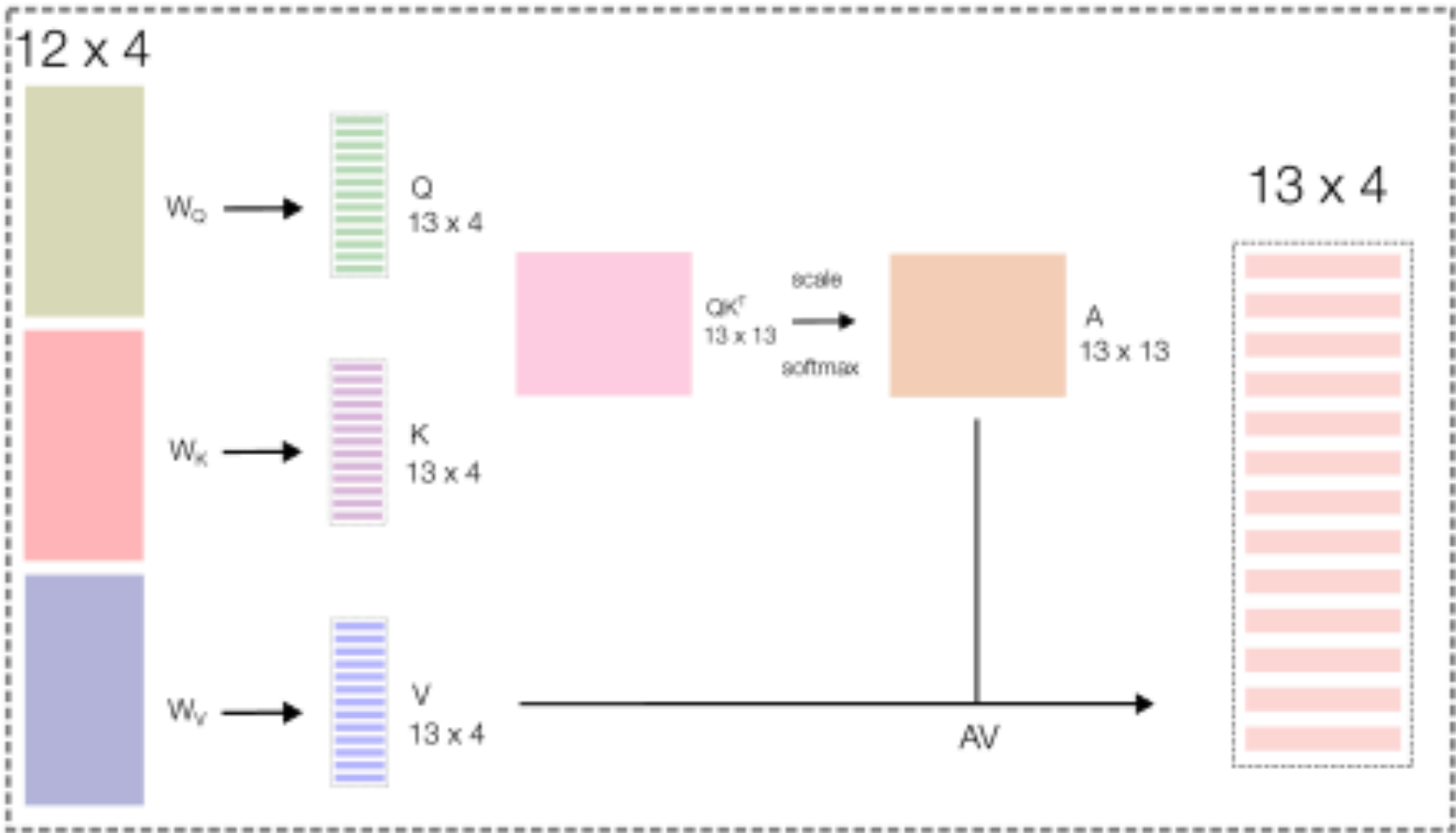
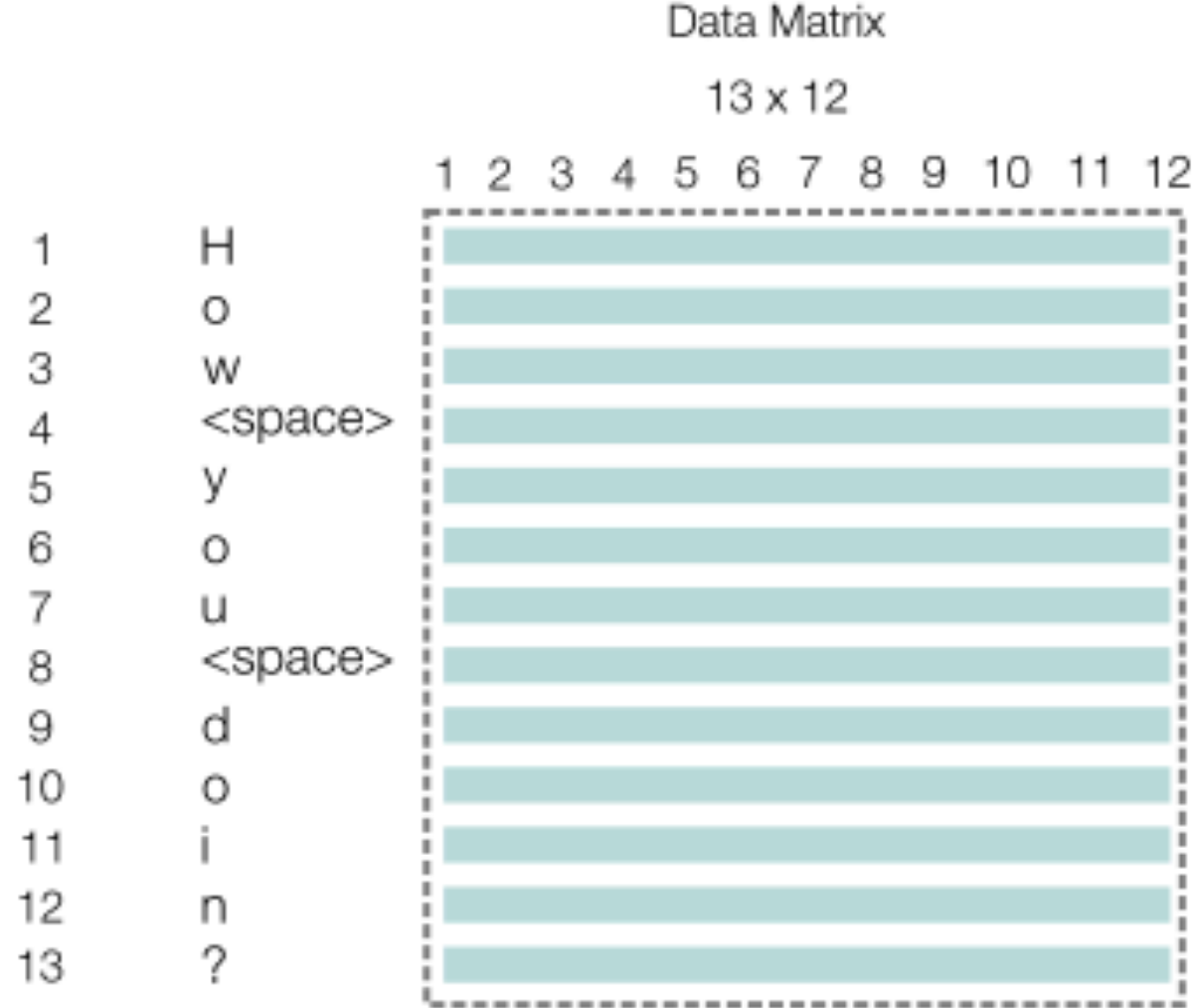
Multi-head Attention



****Multi-head Attention****



****Multi-head Attention****



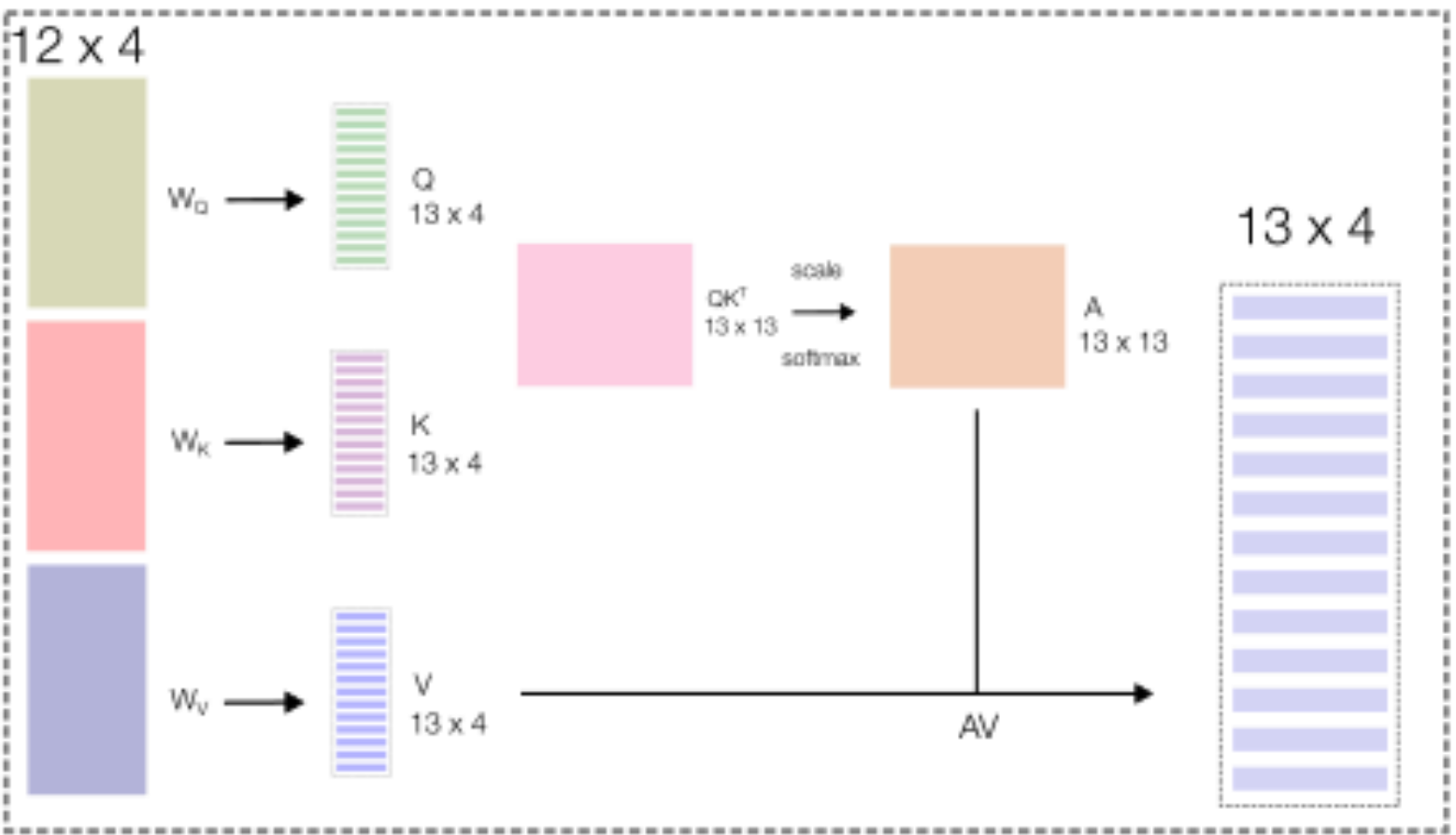
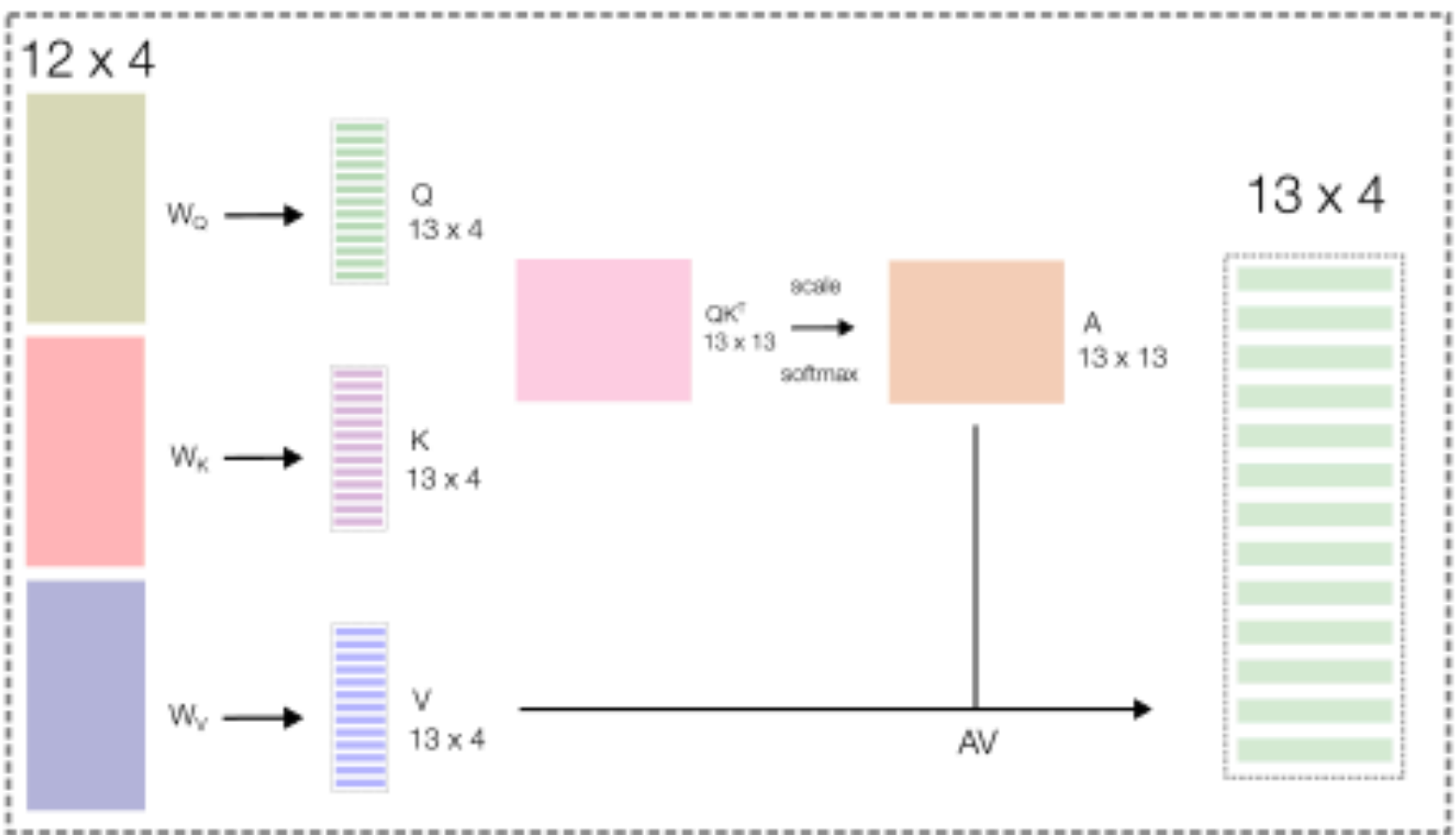
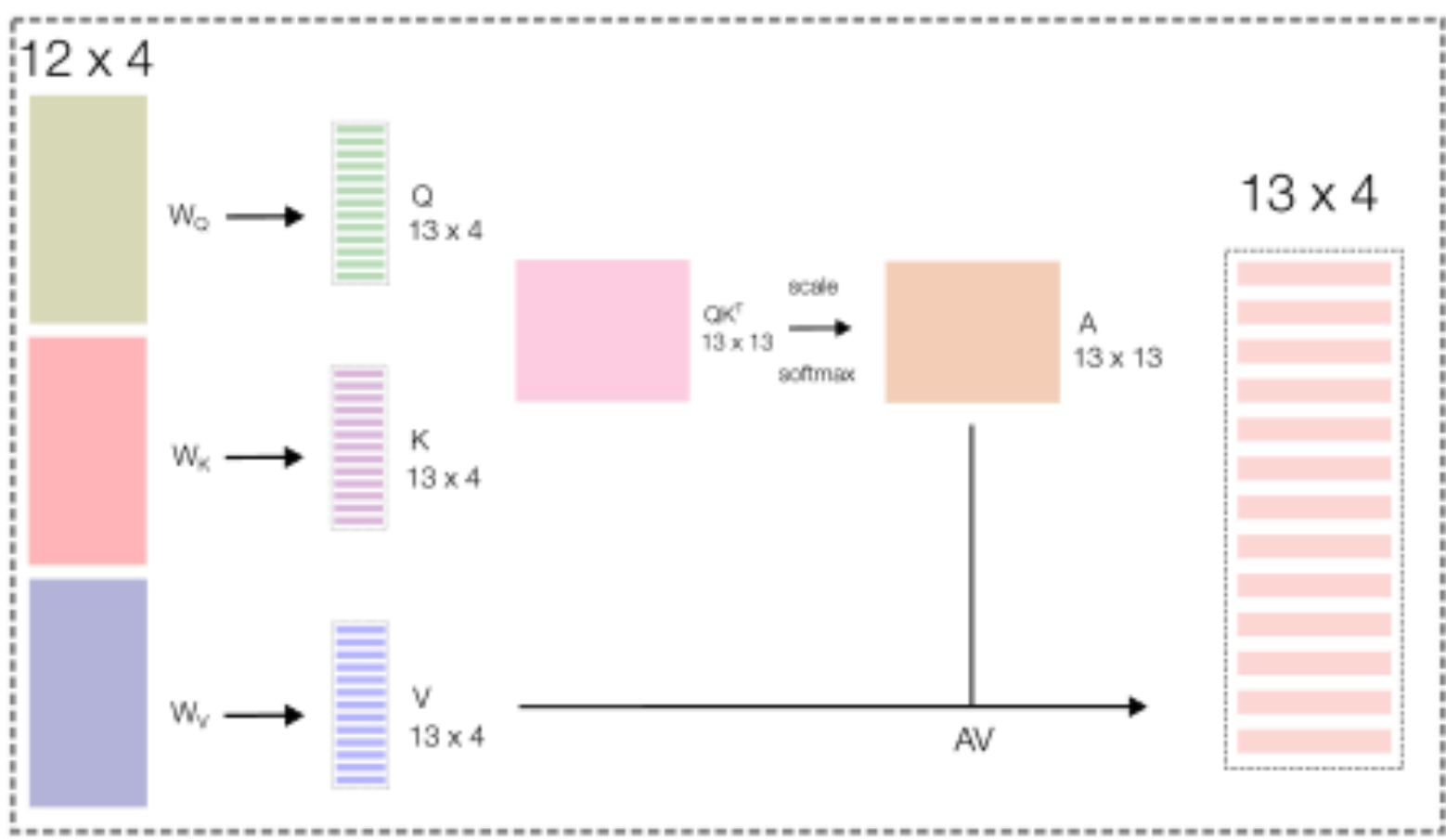
****Multi-head Attention****

1 2 3 4 5 6 7 8 9 10 11 12 13

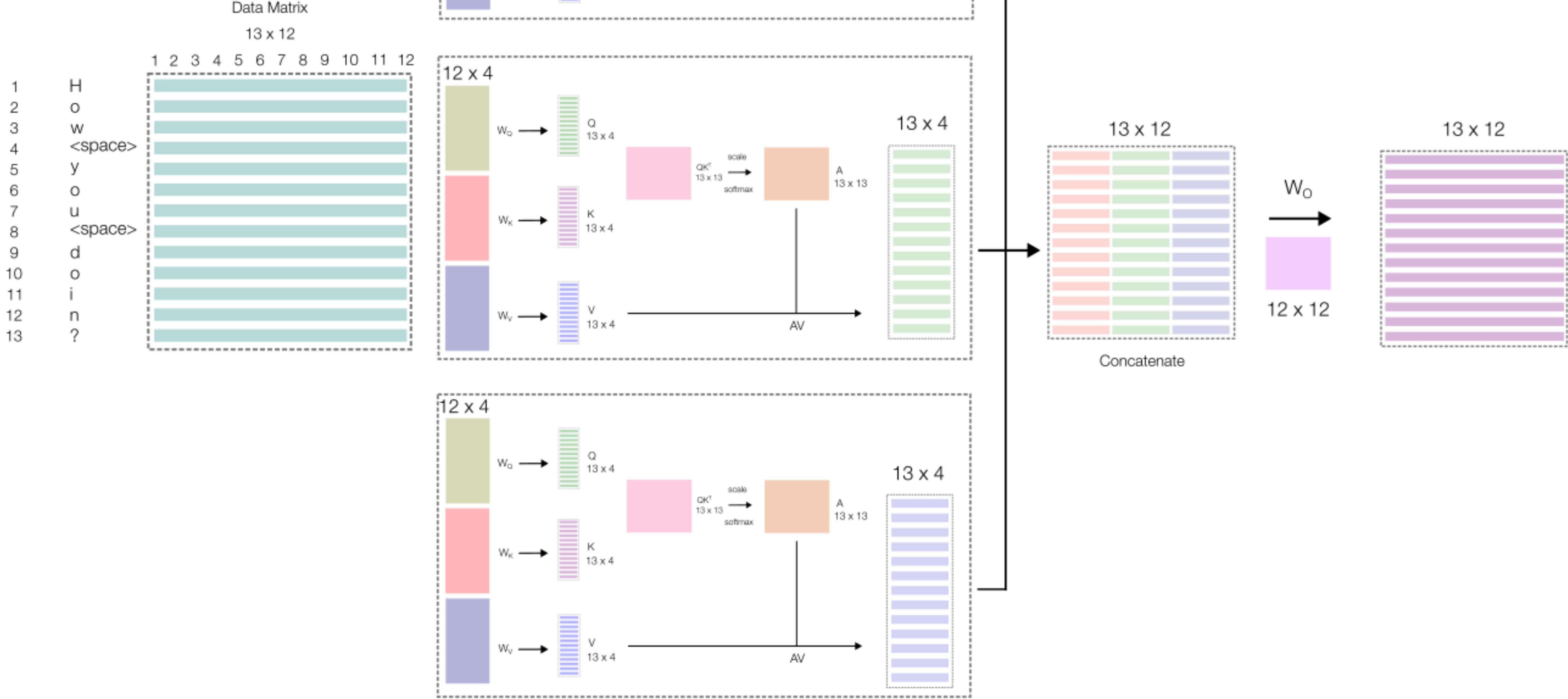
H
o
w
<space>
y
o
u
<space>
d
o
i
n
?

1	2	3	4	5	6	7	8	9	10	11	12

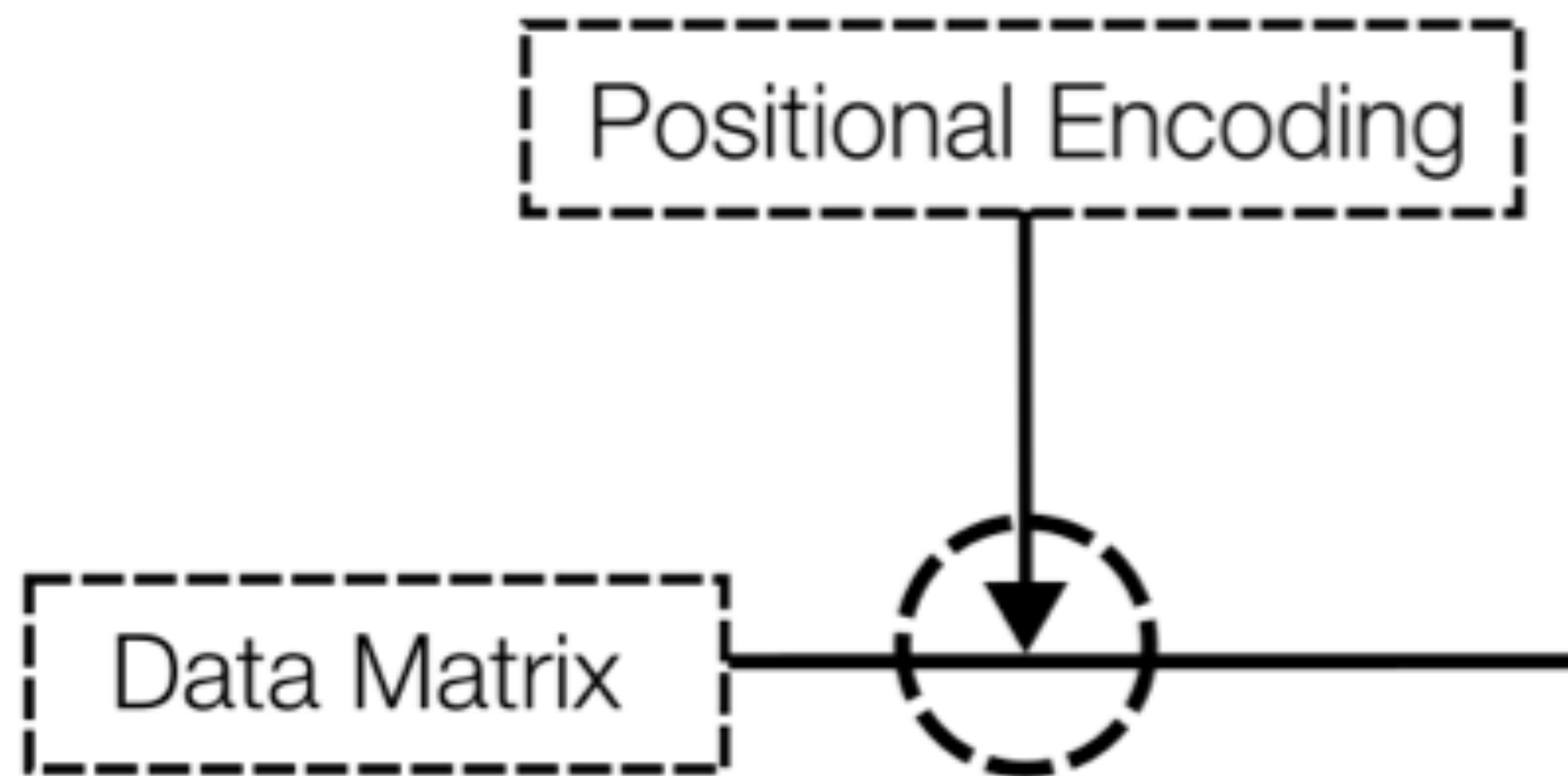
Data Matrix
13 x 12



****Multi-head Attention****

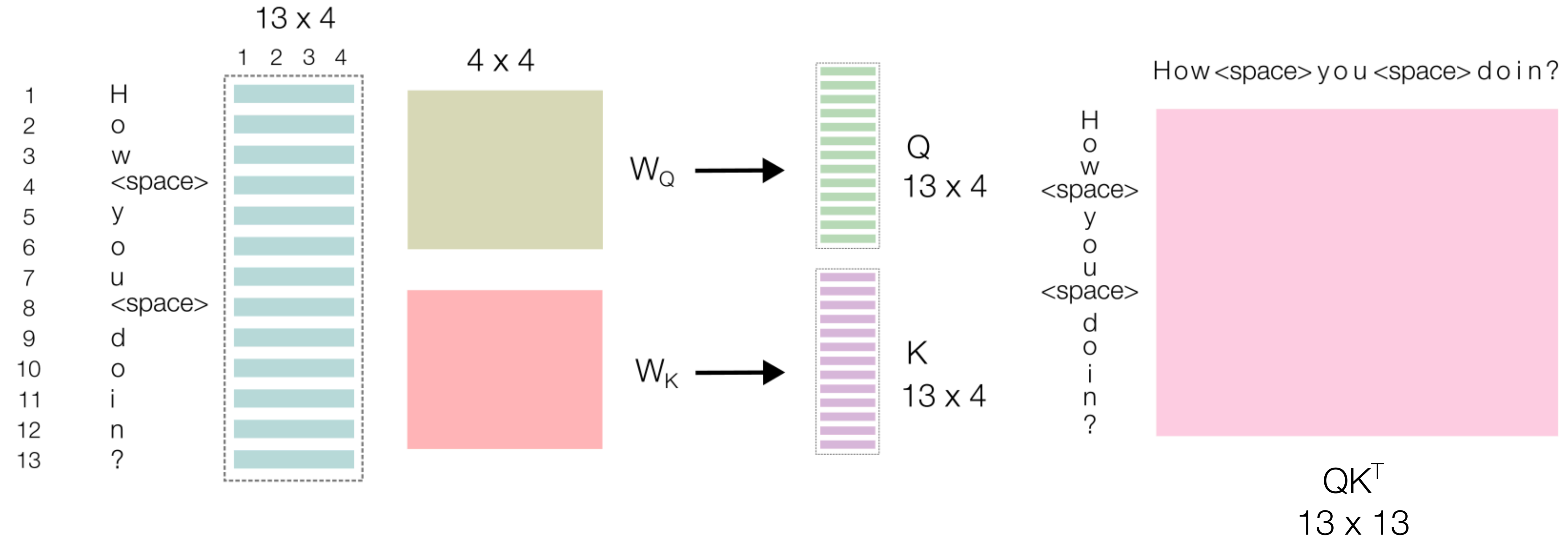


****Attention *as such* is order-agnostic****

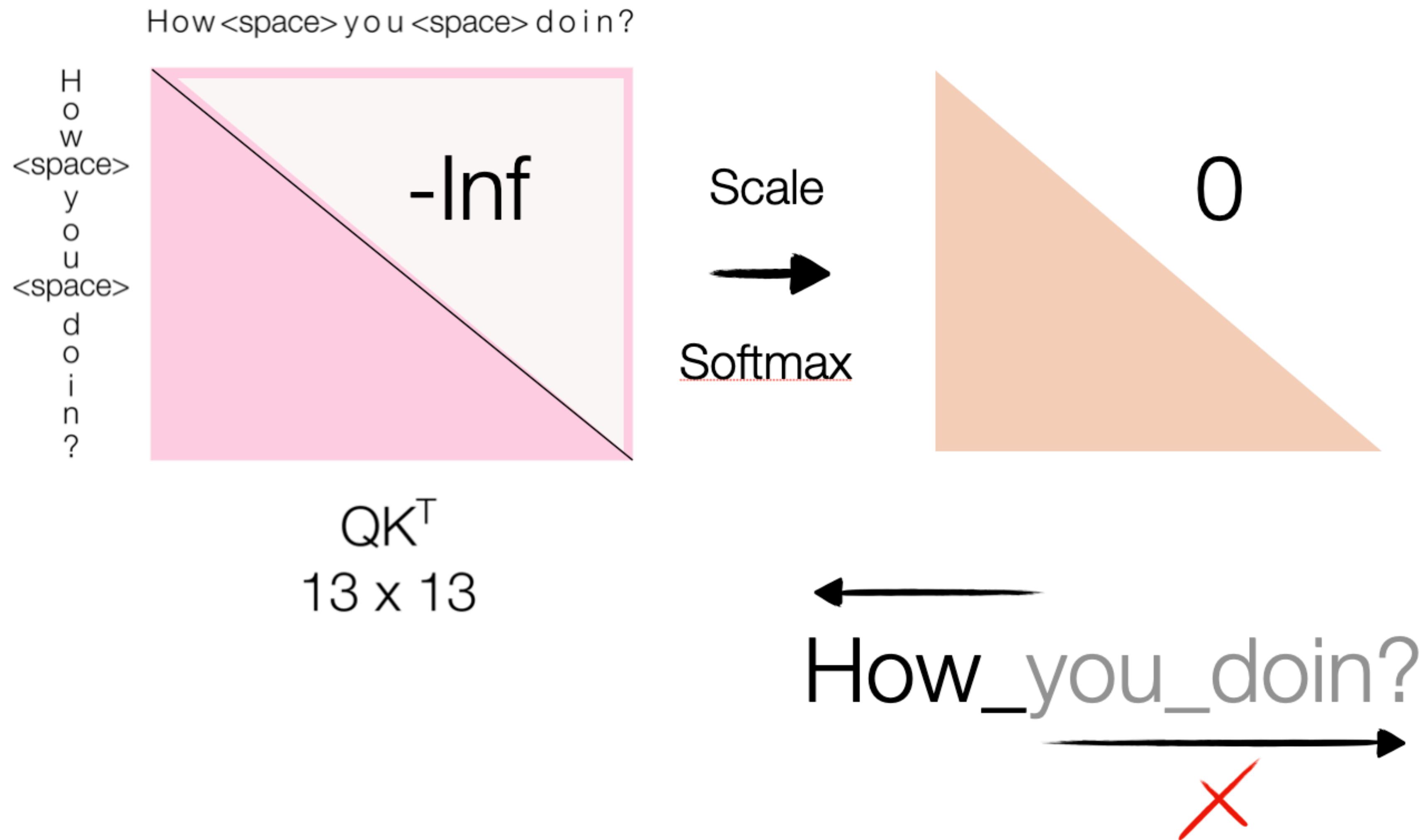


Positional encoding allows token order to affect final representations...

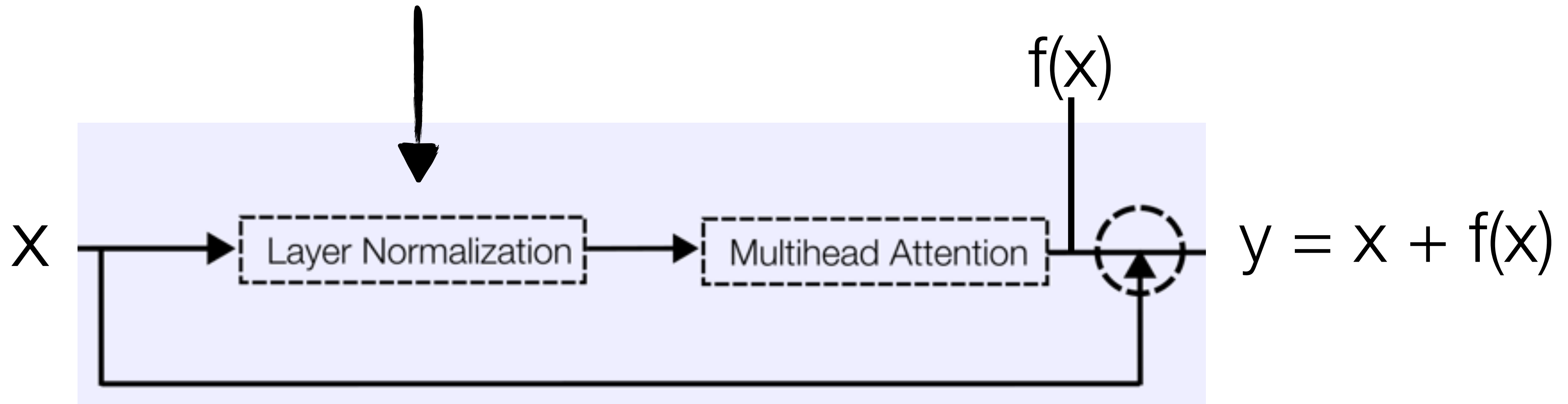
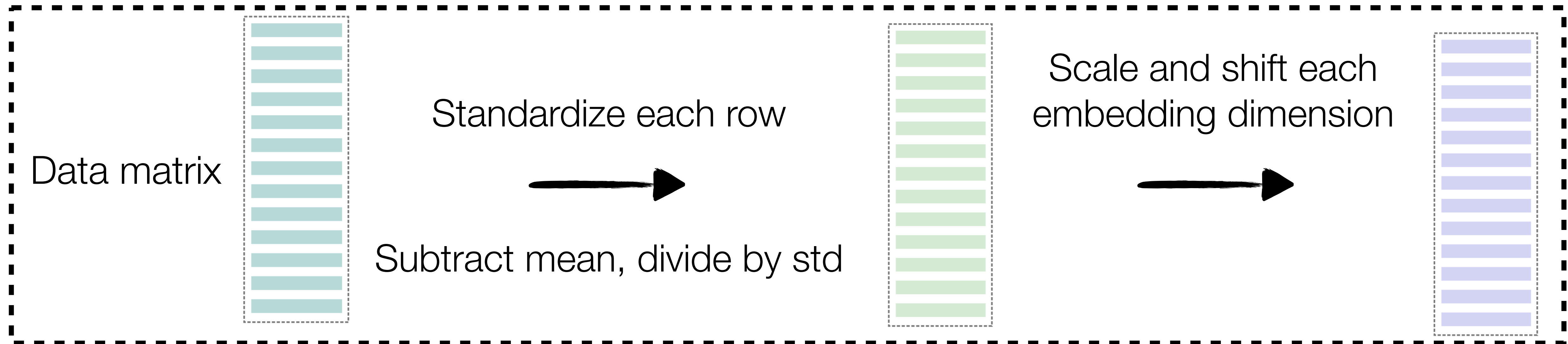
Nuances relating to Next-token prediction: Causal Masking



****Causal Masking****



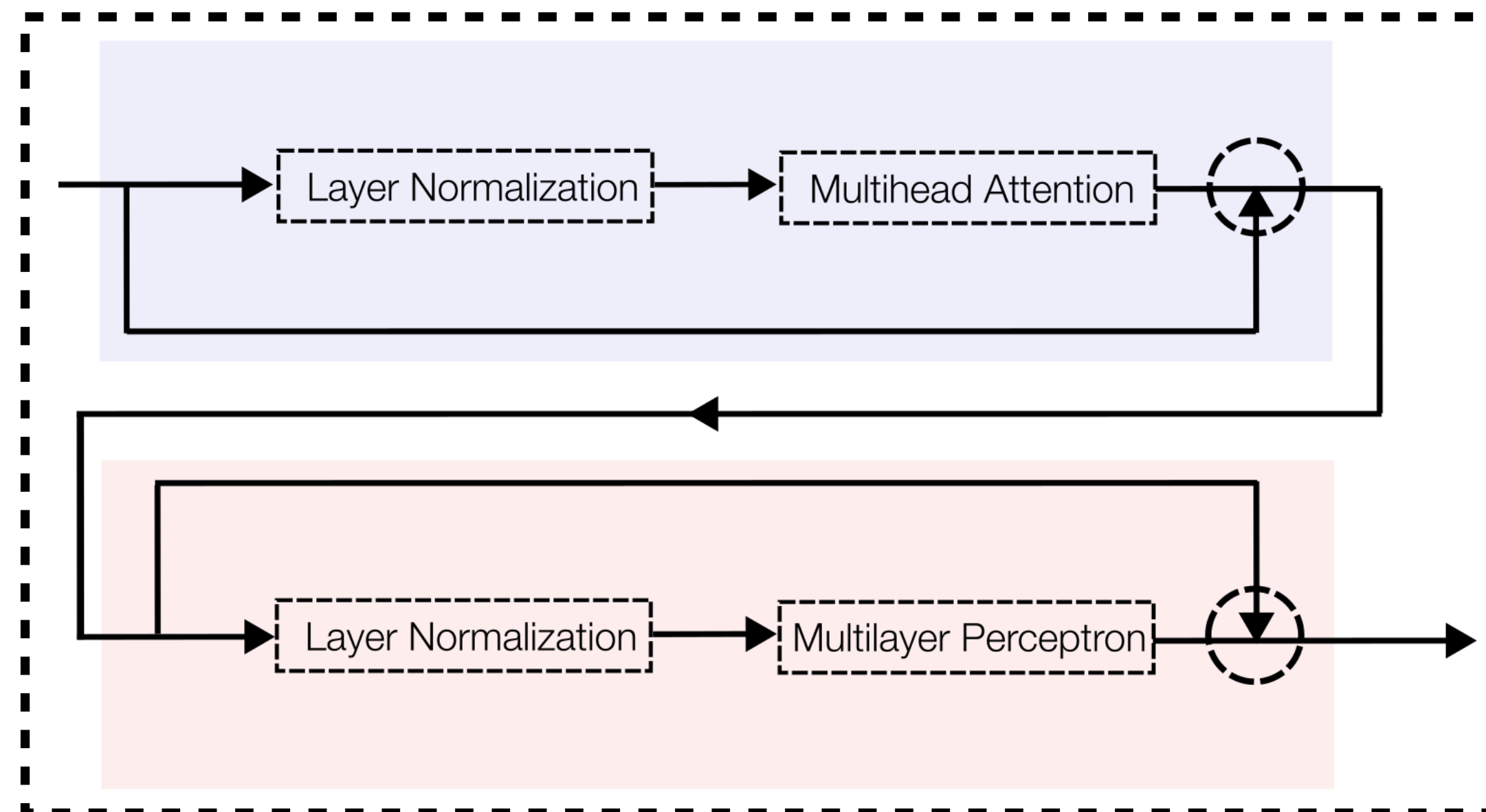
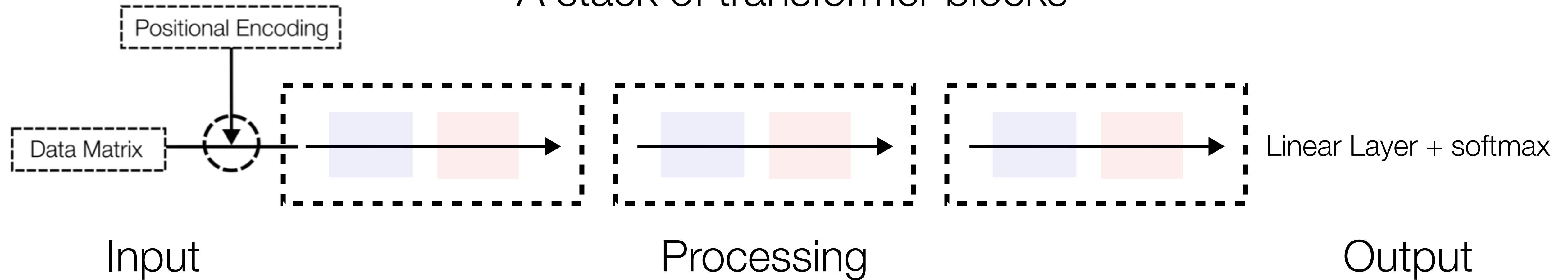
Remaining primitives: Layer Norm and residual connections



Residual connection

Language Model

A stack of transformer blocks



Let's train the language model now!

** Check Colab notebook to train your transformer **

Detailed overview of transformers + code

- Dataset

- Training objective

- Architecture + Training

Progress Bar

- ~~Very short intro to machine learning~~
- ~~Detailed overview of transformers + code~~